



SAPIENZA  
UNIVERSITÀ DI ROMA

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# Modellazione

## MIP

### Numerical problems

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Fifteen numerical models, from EX 1 to EX 15: explicit data, few variables, one step for each technique. They are the easiest examples of the course and give a concrete feel for each modelling technique before the general problems with a symbolic model. For each one: statement, model, instance, a feasible solution, the dual of the relaxation and the optimum, with the script that reproduces them.

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Online version and downloads:

[fabiofurini.github.io/modellazione-mip/](https://fabiofurini.github.io/modellazione-mip/)  
[github.com/fabiofurini/modellazione-mip](https://github.com/fabiofurini/modellazione-mip)

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# Fifteen numerical models

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These notes collect the fifteen numerical models of the course, from EX 1 to EX 15. Explicit data, few variables, one technique per model: they are the easiest ones, and they come before the three families of problems and the mixed problems.

Every numerical model always has the same five parts:

- the statement, with the data of the instance;
- the **variables**, with their domain and their count;
- the **model of the instance**, primal and dual, in the violet box;
- a feasible solution built by hand, which gives the primal bound;
- a dual solution built by hand, which gives the dual bound, and the comparison with the solver optimum.

Every model has its own script and its own notebook, one per model, listed at the beginning of each section.

## EX 1 — The eight-seat van

Script: `python/ex01_van.py`

[Notebook in Colab](#)

### EX 1 — which groups of tourists to accept

A tour operator rents an eight-seat van to guide tourists around a scenic location. On a given day four groups show up, of 2, 3, 4 and 5 people. Since the van cannot take all of them at once, the operator asks each group (whose members do not want to be separated) to make an offer: the four groups offer 30, 50, 80 and 70 euros. For operational reasons at most two groups can be accepted; moreover, if group 2 is accepted then group 4 must be accepted as well. The goal is to maximise the total earnings.

### The model.

With  $x_j = 1$  if group  $j$  is accepted:

## EX 1 on the instance — primal

$$\begin{aligned}
& \max 30x_1 + 50x_2 + 80x_3 + 70x_4 \\
& \text{subject to } 2x_1 + 3x_2 + 4x_3 + 5x_4 \leq 8 \\
& \quad x_1 + x_2 + x_3 + x_4 \leq 2 \\
& \quad \quad x_2 - x_4 \leq 0 \\
& \quad x_1, \quad x_2, \quad x_3, \quad x_4 \in \{0, 1\}
\end{aligned}$$

**Description.** Three techniques of chapter “Links between variables” in four lines: the capacity (section 6.1), the count (6.4) and the logical precedence (6.9). The third constraint reads  $x_2 \leq x_4$ : if  $x_2 = 1$  then  $x_4 = 1$ . The other direction is false and must not hold: group 4 can be accepted without group 2.

**Heuristic: the primal bound.**

Constructive heuristic on the decreasing offer, accepting a group only if all the constraints stay satisfied (the implication included).

- Group 3 (offer 80): accepted, 4 seats, 1 group.
- Group 4 (offer 70): rejected, 9 seats would be needed out of 8.
- Group 2 (offer 50): rejected, it would force accepting group 4.
- Group 1 (offer 30): accepted, 6 seats, 2 groups.

$$z(MILP) \geq LB = 110.$$

**Dual: the dual bound.**

With  $\alpha \geq 0$  on the seats,  $\beta \geq 0$  on the number of groups and  $\gamma \geq 0$  on the implication, the dual is

$$\min 8\alpha + 2\beta \quad \text{s.t.} \quad a_j \alpha + \beta (+\gamma \text{ if } j = 2, -\gamma \text{ if } j = 4) \geq p_j.$$

The recipe:  $\beta = \gamma = 0$  and only the seat is priced, at the price no group can beat,

$$\bar{\alpha} = \max_j \frac{p_j}{a_j} = \max\left(\frac{30}{2}, \frac{50}{3}, \frac{80}{4}, \frac{70}{5}\right) = 20, \quad UB = 8 \cdot 20 = 160.$$

## EX 1 on the instance — dual of the LP relaxation

$$\begin{aligned}
& \min 8\alpha + 2\beta \\
& \text{subject to } 2\alpha + \beta \geq 30 \\
& \quad 3\alpha + \beta + \gamma \geq 50 \\
& \quad 4\alpha + \beta \geq 80 \\
& \quad 5\alpha + \beta - \gamma \geq 70 \\
& \quad \alpha \geq 0 \\
& \quad \beta \geq 0 \\
& \quad \gamma \geq 0
\end{aligned}$$

## Results.

|                | $LB$ | $UB$ | $z(LP)$ | $z(LP^+)$ | $z(MILP)$ |
|----------------|------|------|---------|-----------|-----------|
| EX 1 van (max) | 110  | 160  | 160     | 140       | 120       |

The optimum accepts groups 2 and 4, filling exactly the eight seats, for earnings of 120. The hand-built dual bound is *optimal* for the relaxation without the bounds (160); the strengthened one, which adds  $x_j \leq 1$ , drops to 140 and gets closer to the optimum. Removing the implication the optimum would rise to 130 (groups 2 and 3): the implication costs 10 euros. Removing the limit of two groups changes nothing, because it is the seat capacity that bites.

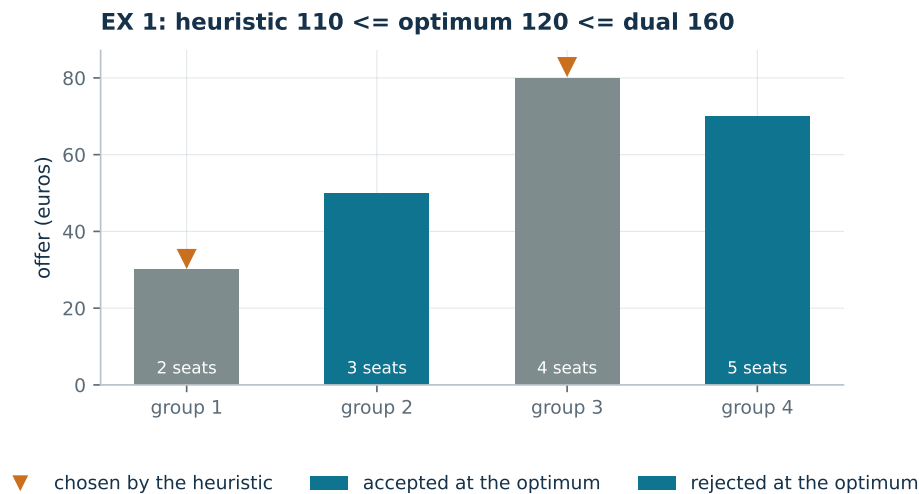


Figure 1: EX 1: the three bounds side by side, from the heuristic to the optimum to the dual built by hand.

## EX 2 — Bus lines

Script: `python/ex02_buslines.py`

[Notebook in Colab](#)

### EX 2 — assignment with a capacity in number of lines

A city council is evaluating the offers of three bus companies for four bus lines. Each company has stated the cost at which it would operate each line:

| Company | Line 1 | Line 2 | Line 3 | Line 4 |
|---------|--------|--------|--------|--------|
| 1       | 10     | 4      | 9      | 7      |
| 2       | 1      | 2      | 3      | 10     |
| 3       | 8      | 9      | 10     | 1      |

Every line must be assigned to exactly one company and every company may operate at most two lines. The total cost is to be minimised.

### The model.

With  $x_{ij} = 1$  if company  $i$  operates line  $j$  ( $3 \cdot 4 = 12$  binary variables):

## EX 2 on the instance — primal

$$\begin{aligned}
& \min 10x_{11} + 4x_{12} + 9x_{13} + 7x_{14} + x_{21} + 2x_{22} + 3x_{23} + 10x_{24} + 8x_{31} + 9x_{32} + 10x_{33} + x_{34} \\
& \text{subject to} \quad x_{11} \qquad \qquad \qquad + x_{21} \qquad \qquad \qquad + x_{31} \qquad \qquad \qquad = 1 \\
& \qquad \qquad \qquad x_{12} \qquad \qquad \qquad + x_{22} \qquad \qquad \qquad + x_{32} \qquad \qquad \qquad = 1 \\
& \qquad \qquad \qquad \qquad x_{13} \qquad \qquad \qquad + x_{23} \qquad \qquad \qquad + x_{33} \qquad \qquad \qquad = 1 \\
& \qquad \qquad \qquad \qquad \qquad x_{14} \qquad \qquad \qquad + x_{24} \qquad \qquad \qquad + x_{34} = 1 \\
& \qquad \qquad \qquad x_{11} + x_{12} + x_{13} + x_{14} \qquad \qquad \qquad \leq 2 \\
& \qquad \qquad \qquad \qquad \qquad x_{21} + x_{22} + x_{23} + x_{24} \qquad \qquad \leq 2 \\
& \qquad \qquad \qquad \qquad \qquad \qquad \qquad x_{31} + x_{32} + x_{33} + x_{34} \leq 2 \\
& \qquad \qquad \qquad x_{11}, \quad x_{12}, \quad x_{13}, \quad x_{14}, \quad x_{21}, \quad x_{22}, \quad x_{23}, \quad x_{24}, \quad x_{31}, \quad x_{32}, \quad x_{33}, \quad x_{34} \in \{0, 1\}
\end{aligned}$$

**Description.** With  $n = 3$  companies,  $m = 4$  lines and  $p = 2$ : four *set partitioning* constraints (one per line) and three capacity constraints (one per company). It is problem 7.1 with the capacity counted in *number* of jobs instead of time.

**Heuristic.**

Constructive heuristic on the lines, in order: every line goes to the cheapest company among those not yet full.

- **Line 1:** costs 10, 1, 8; the cheapest is company 2.
- **Line 2:** costs 4, 2, 9; company 2 again, which is now full.
- **Line 3:** companies 1 and 3 remain, costs 9 and 10: company 1.
- **Line 4:** companies 1 and 3 remain, costs 7 and 1: company 3.

Value  $1 + 2 + 9 + 1 = 13$ , so  $UB = 13$ .

**Dual and lower bound.**

Relaxing  $x_{ij} \in \{0, 1\}$  to  $x_{ij} \geq 0$ , with  $\alpha_j$  free for every line constraint and  $\pi_i \leq 0$  for every capacity constraint:

**Description.** The variable  $\alpha_j$  is the value of line  $j$ , which has to be assigned anyway, and  $\pi_i$  the (non-positive) price of one of the  $p$  lines company  $i$  may run. The objective prices the lines and the available slots. The dual constraints are the columns of the  $x_{ij}$ : giving line  $j$  to company  $i$  satisfies its assignment constraint and takes one slot, and the balance cannot exceed the cost  $c_{ij}$  of that pairing.

With the "zero out and saturate" recipe,  $\bar{\pi} = 0$  and  $\bar{\alpha}_j = \min_i c_{ij} = (1, 2, 3, 1)$ , feasible by construction:  $LB = 7$ . Meaning: "every line costs at least the lowest offer it received".

## EX 2 on the instance — dual of the LP relaxation

$$\begin{aligned}
& \max \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + 2\beta_1 + 2\beta_2 + 2\beta_3 \\
& \text{subject to } \alpha_1 + \beta_1 \leq 10 \\
& \quad \alpha_2 + \beta_1 \leq 4 \\
& \quad \alpha_3 + \beta_1 \leq 9 \\
& \quad \alpha_4 + \beta_1 \leq 7 \\
& \quad \alpha_1 + \beta_2 \leq 1 \\
& \quad \alpha_2 + \beta_2 \leq 2 \\
& \quad \alpha_3 + \beta_2 \leq 3 \\
& \quad \alpha_4 + \beta_2 \leq 10 \\
& \quad \alpha_1 + \beta_3 \leq 8 \\
& \quad \alpha_2 + \beta_3 \leq 9 \\
& \quad \alpha_3 + \beta_3 \leq 10 \\
& \quad \alpha_4 + \beta_3 \leq 1 \\
& \alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0 \\
& \beta_1, \beta_2, \beta_3 \leq 0
\end{aligned}$$

## The bounds.

The optimum is  $z(MILP) = 9$  (lines 1 and 3 to company 2, line 2 to company 1, line 4 to company 3) and both relaxations coincide with it.

| $UB$ (constructive heuristic) | $LB$ (dual by hand) | $z(LP)$ | $z(MILP)$ | heuristic gap |
|-------------------------------|---------------------|---------|-----------|---------------|
| 13                            | 7                   | 9       | 9         | 44.4%         |

Worth noting:  $z(LP) = z(MILP) = 9$ , yet the hand-built bound stops at 7 and the heuristic is off by 44%. The relaxation is exact, the dual recipe is not: they are two different things.

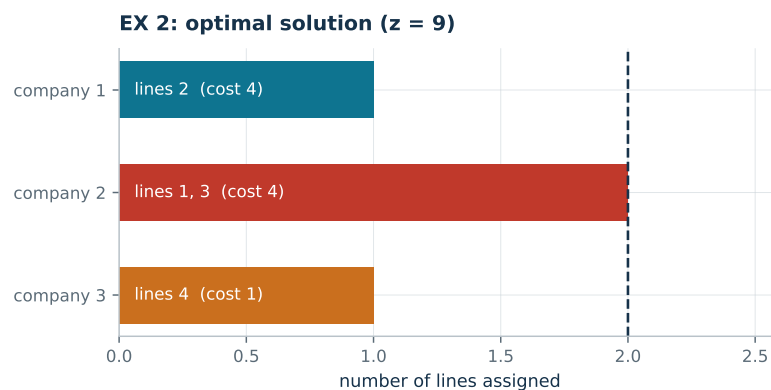


Figure 2: EX 2: the optimal solution. No company exceeds two lines.

## EX 3 — Relay

Script: `python/ex03_relay.py`

[Notebook in Colab](#)

**EX 3 — assignment with more resources than tasks**

A coach must form a relay team by choosing three swimmers out of four. The race has three 50-metre legs: backstroke, breaststroke and butterfly. The recorded times (seconds) are:

| Stroke       | Carlo | Paolo | Fabio | Marco |
|--------------|-------|-------|-------|-------|
| Backstroke   | 37    | 32    | 33    | 37    |
| Breaststroke | 43    | 33    | 42    | 34    |
| Butterfly    | 33    | 28    | 38    | 30    |

The total relay time is to be minimised.

**The model.**

With  $x_{sn} = 1$  if stroke  $s$  is swum by  $n$ :  $\sum_n x_{sn} = 1$  for every stroke (3 constraints),  $\sum_s x_{sn} \leq 1$  for every swimmer (4 constraints), objective  $\min \sum_{s,n} t_{sn} x_{sn}$ . One swimmer stays out: it is an assignment with more resources than tasks.

**EX 3 on the instance — primal**

$$\begin{aligned}
 &\min 37x_{11} + 32x_{12} + 33x_{13} + 37x_{14} + 43x_{21} + 33x_{22} + 42x_{23} + 34x_{24} + 33x_{31} + 28x_{32} + 38x_{33} + 30x_{34} \\
 \text{subject to } &x_{11} + x_{12} + x_{13} + x_{14} = 1 \\
 &x_{21} + x_{22} + x_{23} + x_{24} = 1 \\
 &x_{31} + x_{32} + x_{33} + x_{34} = 1 \\
 &x_{11} + x_{21} + x_{31} \leq 1 \\
 &x_{12} + x_{22} + x_{32} \leq 1 \\
 &x_{13} + x_{23} + x_{33} \leq 1 \\
 &x_{14} + x_{24} + x_{34} \leq 1 \\
 &x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34} \in \{0, 1\}
 \end{aligned}$$

**Description.** The objective sums the times of the assignments chosen. The first three rows say that every stroke is swum by exactly one swimmer; the four that follow, that every swimmer swims at most one stroke. It is the structure of the *assignment* problem: one family of equalities on the rows of the matrix and one of inequalities on the columns. The last row declares the twelve variables binary.

**Heuristic.**

Constructive heuristic on the strokes in the given order: backstroke to Paolo (32), breaststroke to Marco (34, the fastest of the three left), butterfly to Carlo (33). Total  $UB = 99$  seconds.

**Dual and lower bound.**

The same structure as EX 2 with  $p = 1$ :  $\bar{\pi} = 0$  and  $\bar{\alpha}_s = \min_n t_{sn} = (32, 33, 28)$ , so  $LB = 93$ . Meaning: "every stroke costs at least the time of its best specialist". The bound is weak exactly when two strokes share the same best specialist — here Paolo is the fastest in both backstroke and butterfly, and indeed the bound is not attainable.

**EX 3 on the instance — dual of the LP relaxation**

$$\begin{array}{llll}
\max & \alpha_1 + \alpha_2 + \alpha_3 + \beta_1 + \beta_2 + \beta_3 + \beta_4 \\
\text{subject to} & \alpha_1 & + \beta_1 & \leq 37 \\
& \alpha_1 & + \beta_2 & \leq 32 \\
& \alpha_1 & + \beta_3 & \leq 33 \\
& \alpha_1 & + \beta_4 & \leq 37 \\
& \alpha_2 & + \beta_1 & \leq 43 \\
& \alpha_2 & + \beta_2 & \leq 33 \\
& \alpha_2 & + \beta_3 & \leq 42 \\
& \alpha_2 & + \beta_4 & \leq 34 \\
& \alpha_3 + \beta_1 & & \leq 33 \\
& \alpha_3 & + \beta_2 & \leq 28 \\
& \alpha_3 & + \beta_3 & \leq 38 \\
& \alpha_3 & + \beta_4 & \leq 30 \\
& \alpha_1, \alpha_2, \alpha_3 & & \geq 0 \\
& \beta_1, \beta_2, \beta_3, \beta_4 & & \leq 0
\end{array}$$

**The bounds.**

The optimum is  $z(MILP) = 95$  seconds: backstroke to Fabio (33), breaststroke to Marco (34), butterfly to Paolo (28); Carlo stays out.

| $UB$ (constructive heuristic) | $LB$ (dual by hand) | $z(LP)$ | $z(MILP)$ | heuristic gap |
|-------------------------------|---------------------|---------|-----------|---------------|
| 99                            | 93                  | 95      | 95        | 4.2%          |

**Why the relaxation is exact here**

The matrix of the model is that of the assignment problem: totally unimodular. All the vertices of the relaxation are integer, so  $z(LP) = z(MILP)$  and integrality costs nothing — the same phenomenon as alldiff in Section “Alldiff and binary expansion”. This stops being true as soon as a capacity constraint with coefficients other than 0 and 1 is added.

**EX 4 — Hub-and-spoke**

Script: `python/ex04_hub.py`

[Notebook in Colab](#)

**EX 4 — the minimum number of hubs**

An airline organises its network with a *hub-and-spoke* system: a hub serves directly every city within 1000 miles. The company operates on eight cities; for each of them, the cities within 1000 miles are:

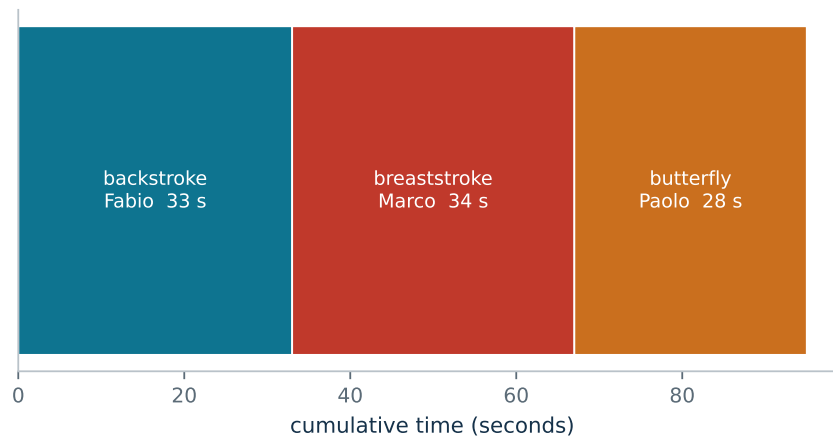
**EX 3: the optimal relay (z = 95 s); Carlo stays out**

Figure 3: EX 3: the optimal relay, leg by leg.

| City          | Cities within 1000 miles            |
|---------------|-------------------------------------|
| Atlanta       | Atlanta, Chicago, Houston, New York |
| Chicago       | Atlanta, Chicago, New York          |
| Denver        | Denver, Los Angeles                 |
| Houston       | Atlanta, Houston                    |
| Los Angeles   | Denver, Los Angeles, San Francisco  |
| New York      | Atlanta, Chicago, New York          |
| San Francisco | Los Angeles, San Francisco, Seattle |
| Seattle       | San Francisco, Seattle              |

The minimum number of hubs is wanted, such that every city is within 1000 miles of at least one hub.

**The model.**

The data are  $n = 8$  cities and  $m = 8$  candidate hubs. With  $y_j = 1$  if city  $j$  becomes a hub ( $m$  binaries) and  $S_i$  the set of cities covering city  $i$ :

**Description.** A pure *set covering*: eight covering constraints, all costs equal to 1. Note that "within 1000 miles" is a symmetric relation, so  $j \in S_i \iff i \in S_j$ : the matrix of the model is symmetric.

## EX 4 on the instance — primal

$$\begin{aligned}
& \min y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 + y_8 \\
& \text{subject to } y_1 + y_2 + y_4 + y_6 \geq 1 \\
& \quad y_1 + y_2 + y_6 \geq 1 \\
& \quad y_3 + y_5 \geq 1 \\
& \quad y_1 + y_4 \geq 1 \\
& \quad y_3 + y_5 + y_7 \geq 1 \\
& \quad y_1 + y_2 + y_6 \geq 1 \\
& \quad y_5 + y_7 + y_8 \geq 1 \\
& \quad y_7 + y_8 \geq 1 \\
& \quad y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8 \in \{0, 1\}
\end{aligned}$$

**Heuristic.**

Constructive covering heuristic (Chapter “Constructive heuristics”), with the criterion "cost per newly covered city".

- **Step 1.** Ratios 1/4 for Atlanta, 1/3 for Chicago, New York and San Francisco, 1/2 for the others: **Atlanta** is chosen, covering Atlanta, Chicago, Houston and New York.
- **Step 2.** Denver, Los Angeles, San Francisco and Seattle remain; ratios 1/2 for Denver, 1/3 for Los Angeles and San Francisco, 1/2 for Seattle: **Los Angeles** is chosen, covering Denver, Los Angeles and San Francisco.
- **Step 3.** Seattle remains; ratio 1/1 for San Francisco and for Seattle: **San Francisco** is chosen.

Three hubs, so  $UB = 3$ .

**Dual and lower bound.**

Relaxing to  $y_j \geq 0$ , with  $\pi_i \geq 0$  for every covering constraint ( $\geq$  direction in a minimisation):

**Description.** The variable  $\pi_i$  is the price attached to city  $i$ . The objective adds up the prices of the eight cities. The constraints are the columns of the  $y_j$ , one per hub: the cities hub  $j$  covers, priced at those values, cannot be worth more than what it costs to open it, namely 1.

Dual constructive heuristic on the cities, in order:  $\pi_i$  is raised until the first opposing dual constraint becomes tight, and the residuals are updated.

- **Atlanta** is covered by four hubs, all with residual 1:  $\bar{\pi}_1 = 1$ , and the residuals of Atlanta, Chicago, Houston and New York drop to zero.
- **Chicago, Houston and New York** have all their hubs at residual zero:  $\bar{\pi} = 0$ .
- **Denver** is covered by Denver and Los Angeles, both with residual 1:  $\bar{\pi}_3 = 1$ , and their residuals drop to zero.
- **Los Angeles and San Francisco** have one hub at residual zero:  $\bar{\pi} = 0$ .

- **Seattle** is covered by San Francisco and Seattle, both with residual 1:  $\bar{\pi}_8 = 1$ .

Value  $LB = 1 + 1 + 1 = 3$ .

#### What this bound proves, in words

The three cities that received  $\pi_i = 1$  — Atlanta, Denver, Seattle — are pairwise "far apart": no city can serve as a hub for two of them. Hence at least three hubs are needed. The dual has formalised an argument one could tell without writing it: that is the point of the constructive heuristic recipe.

#### EX 4 on the instance — dual of the LP relaxation

$$\begin{aligned}
 & \max \pi_1 + \pi_2 + \pi_3 + \pi_4 + \pi_5 + \pi_6 + \pi_7 + \pi_8 \\
 & \text{subject to} \quad \pi_1 + \pi_2 \quad \quad + \pi_4 \quad \quad + \pi_6 \quad \leq 1 \\
 & \quad \quad \quad \pi_1 + \pi_2 \quad \quad \quad + \pi_6 \quad \leq 1 \\
 & \quad \quad \quad \quad \pi_3 \quad \quad + \pi_5 \quad \leq 1 \\
 & \quad \quad \quad \pi_1 \quad \quad + \pi_4 \quad \leq 1 \\
 & \quad \quad \quad \quad \pi_3 \quad + \pi_5 \quad + \pi_7 \leq 1 \\
 & \quad \quad \pi_1 + \pi_2 \quad \quad \quad + \pi_6 \quad \leq 1 \\
 & \quad \quad \quad \quad \quad \pi_5 \quad + \pi_7 + \pi_8 \leq 1 \\
 & \quad \quad \quad \quad \quad \quad \pi_7 + \pi_8 \leq 1 \\
 & \quad \pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6, \pi_7, \pi_8 \geq 0
 \end{aligned}$$

#### The bounds.

$z(MILP) = 3$ , with hubs in Atlanta, Los Angeles and San Francisco — the same solution as the constructive heuristic.

| $UB$ (constructive heuristic) | $LB$ (dual by hand) | $z(LP)$ | $z(MILP)$ | heuristic gap |
|-------------------------------|---------------------|---------|-----------|---------------|
| 3                             | 3                   | 3       | 3         | 0.0%          |

The two hand-built bounds coincide: optimality is proved without the solver.

## EX 5 — Mutual funds bought in lots

Script: `python/ex05_funds.py`

[Notebook in Colab](#)

#### EX 5 — four mutual funds

After a lottery win, an investor must decide how to allocate 100 million lire. A friend suggests four types of mutual fund. Shares can only be bought in indivisible lots, costing 12, 15, 18 and 25 million respectively. The estimated annual return is  $1/6$  of the capital invested for the first fund,  $1/3$  for the second,  $1/9$  for the third and 16% for the fourth. Moreover the number of lots of the second fund cannot exceed 50% of the total number of lots bought. The goal is to maximise the expected annual return.

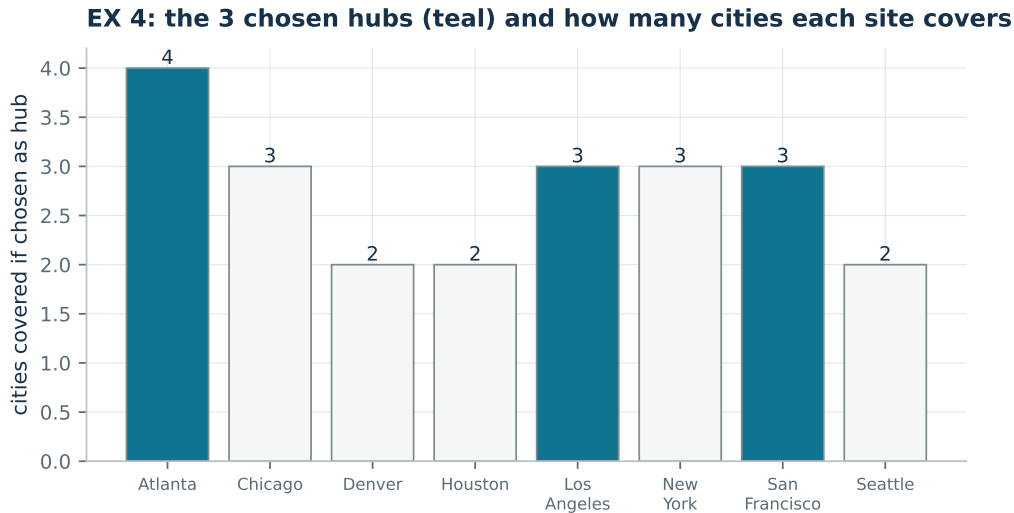


Figure 4: EX 4: for every city, how many it would cover as a hub; in teal the three chosen ones.

### The model.

The return of one lot is  $12 \cdot 1/6 = 2$  million for the first fund,  $15 \cdot 1/3 = 5$  for the second,  $18 \cdot 1/9 = 2$  for the third and  $25 \cdot 0.16 = 4$  for the fourth. The share constraint  $x_2 \leq \frac{1}{2}(x_1 + x_2 + x_3 + x_4)$ , multiplied by 2 and moved to the left, becomes  $-x_1 + x_2 - x_3 - x_4 \leq 0$ . With  $x_j$  the number of lots of fund  $j$ :

#### EX 5 — the model written out in full

$$\begin{aligned}
 &\max \quad 2x_1 + 5x_2 + 2x_3 + 4x_4 \\
 &\text{subject to} \quad 12x_1 + 15x_2 + 18x_3 + 25x_4 \leq 100 \\
 &\quad \quad \quad -x_1 + x_2 - x_3 - x_4 \leq 0 \\
 &\quad \quad \quad x_1, \quad x_2, \quad x_3, \quad x_4 \in \mathbb{Z}_{\geq 0}
 \end{aligned}$$

**Description.** An integer, non-binary knapsack: the variables count lots, not choices. The first row is the budget, the second the share of the second fund written in linear form; the last declares the four variables integer and non-negative.

### Heuristic: the primal bound.

Constructive heuristic on the return per million invested:  $1/3$  for the second fund,  $1/6$  for the first,  $4/25$  for the fourth and  $1/9$  for the third. One starts from the second, the best one, and buys no lot of it at all: on its own it would break the share, which grants it at most half of the total. One moves to the first, buys 8 lots (96 million) and is left with 4 million, not enough for any other lot. The heuristic has tripped over the best fund: buying a lot of it pays, but only *together* with a lot of another fund, and a rule that looks at one element at a time cannot see that.

$$z(MILP) \geq LB = 16.$$

### Dual: the dual bound.

With  $\alpha \geq 0$  on the budget and  $\beta \geq 0$  on the share, one row per fund:

## EX 5 on the instance — dual

$$\begin{aligned}
& \min 100\alpha \\
& \text{subject to } 12\alpha - \beta \geq 2 \\
& \quad 15\alpha + \beta \geq 5 \\
& \quad 18\alpha - \beta \geq 2 \\
& \quad 25\alpha - \beta \geq 4 \\
& \quad \alpha \geq 0 \\
& \quad \beta \geq 0
\end{aligned}$$

A choice that comes naturally is  $\bar{\alpha} = 5/32$ ,  $\bar{\beta} = 1/8$ . Let us check the first constraint:

$$12 \cdot \frac{5}{32} - \frac{1}{8} = \frac{60}{32} - \frac{4}{32} = \frac{56}{32} = \frac{7}{4} < 2.$$

It is violated by 1/4: that pair is *not* a feasible dual solution, and the value  $100 \cdot 5/32$  is not a bound. A dual value may be read as a bound only after checking *all* the constraints, one by one.

The correct recipe is  $\bar{\beta} = 0$  and

$$\bar{\alpha} = \max_j \frac{p_j}{c_j} = \max\left(\frac{2}{12}, \frac{5}{15}, \frac{2}{18}, \frac{4}{25}\right) = \frac{1}{3},$$

that is, a million is worth what it returns in the best fund. All the constraints become  $c_j \bar{\alpha} \geq p_j$  and are satisfied, and

$$z(MILP) \leq UB = 100 \cdot \frac{1}{3} = \frac{100}{3} \approx 33.33.$$

## Results.

|                  | $LB$ | $UB$  | $z(LP)$ | $z(LP^+)$ | $z(MILP)$ |
|------------------|------|-------|---------|-----------|-----------|
| EX 5 funds (max) | 16   | 100/3 | 700/27  | 700/27    | 23        |

The optimum buys 4 lots of the first fund and 3 of the second, 93 million for a return of 23: the heuristic stops at 16 and is 30% off. The difference  $700/27 - 23 = 79/27$  is the price of integrality: the relaxation would buy lots in pieces. The dual bound built by hand,  $100/3$ , stays far away: "a million is worth what it returns in the best fund" ignores that the best fund cannot be bought on its own. It is a valid recipe, not a good one.

Dropping the share constraint the optimum would rise from 23 to 30: six lots of the second fund and nothing else. The share therefore costs 7 million of return a year, and this is how one measures what a constraint really does: remove it and see how much the optimum moves.

## EX 6 — Vehicles with a minimum quantity

Script: `python/ex06_vehicles.py`

[Notebook in Colab](#)

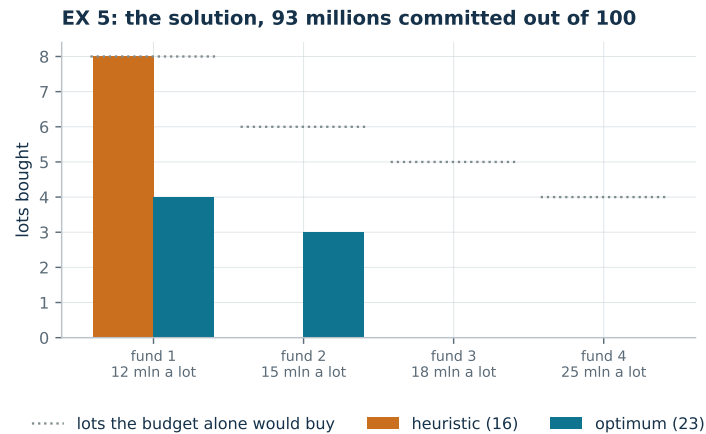


Figure 5: EX 5: the lots bought by the heuristic and those bought at the optimum, with the number of lots the budget alone would allow for each fund.

### EX 6 — five types of vehicle

A company is considering the production of five vehicle types: compact car, midsize car, large car, midsize minivan and large minivan. It has 1000 tons of steel and 2000 labour hours available. The table reports, for every type, the steel per unit, the hours per unit, the unit profit and the minimum quantity to be produced if the type is selected:

| Type            | steel | hours | profit | minimum |
|-----------------|-------|-------|--------|---------|
| compact car     | 2     | 30    | 200    | 10      |
| midsize car     | 3     | 25    | 250    | 10      |
| large car       | 5     | 40    | 300    | 10      |
| midsize minivan | 6     | 45    | 550    | 5       |
| large minivan   | 8     | 55    | 700    | 5       |

The company wants the plan of maximum profit.

### The model.

It is model 9.3 without the bonus: the same semi-integer variables, with  $M_j = \min\{\lfloor b_i/a_{ij} \rfloor : a_{ij} > 0\}$ . On the instance  $M = (66, 80, 50, 44, 36)$ , and in every case the limiting resource is the labour time.

**Description.** The minimum-quantity constraint and the activation constraint together make  $x_j$  a *semi-integer* variable — the units are whole numbers — as the truth table shows:

| $y_j$ | what the two constraints impose   | $x_j$ allowed           |
|-------|-----------------------------------|-------------------------|
| 0     | $x_j \geq 0$ and $x_j \leq 0$     | only $x_j = 0$          |
| 1     | $x_j \geq q_j$ and $x_j \leq M_j$ | $q_j \leq x_j \leq M_j$ |

Neither is enough on its own: without the activation a type could be produced with  $y_j = 0$ ; without the minimum quantity the minimum lot would be empty.

## EX 6 on the instance — primal

$$\begin{array}{llllll}
\max & 200x_1 + 250x_2 + 300x_3 + 550x_4 + 700x_5 \\
\text{subject to} & 2x_1 + 3x_2 + 5x_3 + 6x_4 + 8x_5 & \leq & 1000 \\
& 30x_1 + 25x_2 + 40x_3 + 45x_4 + 55x_5 & \leq & 2000 \\
& x_1 & & -10y_1 & \geq & 0 \\
& & x_2 & & -10y_2 & \geq & 0 \\
& & & x_3 & & -10y_3 & \geq & 0 \\
& & & & x_4 & & -5y_4 & \geq & 0 \\
& & & & & x_5 & & -5y_5 & \geq & 0 \\
& x_1 & & & & -66y_1 & \leq & 0 \\
& & x_2 & & & -80y_2 & \leq & 0 \\
& & & x_3 & & -50y_3 & \leq & 0 \\
& & & & x_4 & & -44y_4 & \leq & 0 \\
& & & & & x_5 & & -36y_5 & \leq & 0 \\
& x_1, & x_2, & x_3, & x_4, & x_5 & & & \in & \mathbb{Z}_{\geq 0} \\
& & & & & & y_1, & y_2, & y_3, & y_4, & y_5 \in \{0, 1\}
\end{array}$$

**Heuristic: the primal bound.**

Constructive heuristic on the profit per labour hour:  $20/3$ ,  $10$ ,  $15/2$ ,  $110/9$  and  $140/11$  for the five types. One starts from the large minivan ( $140/11$ ), produces 36 of them (the maximum allowed by the hours), and is left with 712 tons of steel but only 20 hours: no other type reaches its own minimum quantity and the constructive heuristic stops.

$$z(MILP) \geq LB = 25\,200.$$

**Dual: the dual bound.**

With  $\ell_j = \beta_j = 0$  (the minimum lot is not priced) and a single resource priced at the highest unit price among the types:

$$\begin{aligned}
\bar{\pi}_{\text{steel}} &= \max_j \frac{\bar{p}_j}{a_{1j}} = 100 \Rightarrow 1000 \cdot 100 = 100\,000, \\
\bar{\pi}_{\text{hours}} &= \max_j \frac{\bar{p}_j}{a_{2j}} = \frac{140}{11} \Rightarrow 2000 \cdot \frac{140}{11} = \frac{280\,000}{11} \approx 25\,454.5.
\end{aligned}$$

The better bound comes from the hours:

$$z(MILP) \leq UB = \frac{280\,000}{11}.$$

## EX 6 on the instance — dual

$$\begin{array}{llllllllll}
\min & 1000\pi_1 + 2000\pi_2 & & & & & & & & \\
\text{subject to} & 2\pi_1 & +30\pi_2 & +\lambda_1 & & & +\mu_1 & & & \geq 200 \\
& 3\pi_1 & +25\pi_2 & & +\lambda_2 & & & +\mu_2 & & \geq 250 \\
& 5\pi_1 & +40\pi_2 & & & +\lambda_3 & & & +\mu_3 & \geq 300 \\
& 6\pi_1 & +45\pi_2 & & & & +\lambda_4 & & & +\mu_4 \geq 550 \\
& 8\pi_1 & +55\pi_2 & & & & & +\lambda_5 & & +\mu_5 \geq 700 \\
& & & -10\lambda_1 & & & & -66\mu_1 & & \geq 0 \\
& & & & -10\lambda_2 & & & & -80\mu_2 & \geq 0 \\
& & & & & -10\lambda_3 & & & & -50\mu_3 \geq 0 \\
& & & & & & -5\lambda_4 & & & -44\mu_4 \geq 0 \\
& & & & & & & -5\lambda_5 & & -36\mu_5 \geq 0 \\
& \pi_1, & \pi_2 & & & & & & & \geq 0 \\
& & & \lambda_1, & \lambda_2, & \lambda_3, & \lambda_4, & \lambda_5 & & \leq 0 \\
& & & & & & & & \mu_1, & \mu_2, & \mu_3, & \mu_4, & \mu_5 \geq 0
\end{array}$$

**Description.** The variable  $\pi_i$  is the price of one unit of resource  $i$ , while  $\ell_j$  and  $\mu_j$  are the prices of the two constraints that make  $x_j$  semi-integer. The objective values all the available resources at those prices. The first group of constraints is made of the columns of the  $x_j$ : the resources consumed by one unit of type  $j$ , corrected by the two semi-integrality constraints, must cover the profit  $p_j$ . The second group is made of the columns of the  $y_j$ : switching type  $j$  on forces at least  $q_j$  units and allows at most  $M_j$ , and the balance of the two prices must stay non-negative.

## Results.

|                     | <i>LB</i> | <i>UB</i> | $z(LP)$   | $z(LP^+)$ | $z(MILP)$ |
|---------------------|-----------|-----------|-----------|-----------|-----------|
| EX 6 vehicles (max) | 25 200    | 280000/11 | 280000/11 | 229000/9  | 25 250    |

The optimum produces 9 midsize minivans and 29 large minivans, using all 2000 hours and 286 tons of steel out of 1000. The dual bound is *optimal* for the relaxation without the bounds, and the heuristic is off by a mere 0.2%. Without minimum quantities the optimum would be 25 300: the minimum lot costs 50 euros of profit. Doubling the minimum quantities one drops to 25 200, and doubling the labour hours one reaches 50 750 with the hours still saturated.

## EX 7 — Custom aircraft with a fixed set-up cost

Script: `python/ex07_aircraft.py`[Notebook in Colab](#)

## EX 7 — which orders to accept

A company builds custom-designed aircraft; before producing for a client a fixed set-up cost has to be paid. Three clients have ordered, respectively, 3, 2 and 5 aircraft. The production capacity is 1000 hours. The company decides which orders to accept and how many units to produce for each (possibly fewer than requested). Set-up costs, unit profits and hours per

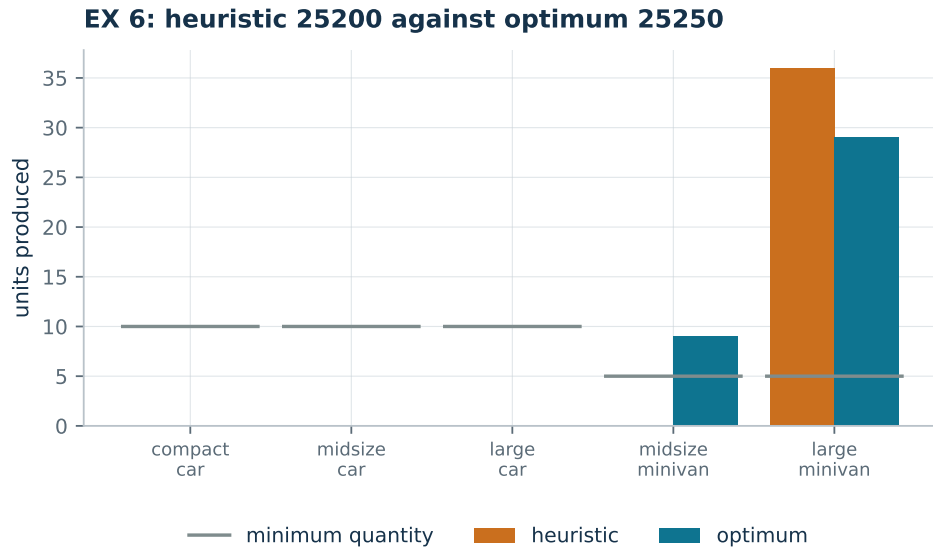


Figure 6: EX 6: the quantities produced by the heuristic and the optimal ones, with the minimum quantity and the maximum producible of each type.

aircraft:

|                        | client 1 | client 2 | client 3 |
|------------------------|----------|----------|----------|
| set-up (millions)      | 3        | 2        | 0        |
| unit profit (millions) | 2        | 3        | 1        |
| hours per aircraft     | 200      | 400      | 200      |

The goal is to maximise the total profit net of the set-up costs.

### The model.

The data are  $n = 3$  clients, with unit profit  $p_j$ , setup cost  $f_j$ , hours per aircraft  $h_j$  and maximum order  $M_j$ , plus the production capacity  $H = 1000$  hours. With  $x_j$  aircraft produced for client  $j$  (integer) and  $y_j$  the indicator of an accepted order:

#### EX 7 on the instance — primal

$$\begin{aligned}
 & \max \quad 2x_1 + 3x_2 + x_3 - 3y_1 - 2y_2 \\
 & \text{subject to} \quad 200x_1 + 400x_2 + 200x_3 \leq 1000 \\
 & \quad x_1 - 3y_1 \leq 0 \\
 & \quad x_2 - 2y_2 \leq 0 \\
 & \quad x_3 - 5y_3 \leq 0 \\
 & \quad x_1, x_2, x_3 \in \mathbb{Z}_{\geq 0} \\
 & \quad y_1, y_2, y_3 \in \{0, 1\}
 \end{aligned}$$

**Description.** A single constraint per client does two things at once: it caps the quantity at the number of aircraft ordered and it charges the set-up. Client 3 has a zero set-up, so  $y_3$  does not appear in the objective and equals 1 in every optimal solution with  $x_3 > 0$ .

**Heuristic: the primal bound.**

Constructive heuristic on the net profit per hour, computed on the whole order:  $3/600$ ,  $4/800$  and  $5/1000$ , that is  $1/200$  for all three. With equal ratios one scans in the natural order: client 1 is accepted with 3 aircraft (600 hours, net profit 3), then client 2 with a single aircraft (400 hours, net profit  $3 - 2 = 1$ ), and the hours run out.

$$z(MILP) \geq LB = 4.$$

**Dual: the dual bound.**

With  $\alpha \geq 0$  on the capacity and  $\beta_j \geq 0$  on the links:

$$\min 1000\alpha \quad \text{s.t.} \quad h_j\alpha + \beta_j \geq p_j, \quad M_j\beta_j \leq f_j.$$

The recipe:  $\bar{\beta}_j = f_j/M_j$ , that is the set-up spread over the aircraft ordered (1, 1 and 0), whence  $\alpha \geq (p_j - \bar{\beta}_j)/h_j$  for every  $j$ , that is  $\alpha \geq 1/200$  in all three cases. With  $\bar{\alpha} = 1/200$ ,

$$z(MILP) \leq UB = 1000 \cdot \frac{1}{200} = 5.$$

**Results.**

|                     | $LB$ | $UB$ | $z(LP)$ | $z(LP^+)$ | $z(MILP)$ |
|---------------------|------|------|---------|-----------|-----------|
| EX 7 aircraft (max) | 4    | 5    | 5       | 5         | 5         |

The optimum is 5 and is attained by *three* different plans: five aircraft for client 3; two for client 2 and one for the third; three for client 1 and two for the third. All three use exactly the 1000 hours. The dual bound is exact, and in this case the certificate settles the problem on its own: there is nothing left for the solver to do. Without set-up costs the optimum would rise to 9; with 1400 hours available, to 7.

**EX 8 — Seminars**

Script: `python/ex08_seminars.py`

[Notebook in Colab](#)

**EX 8 — selection with exact cardinality and non-adjacency**

A student must attend *exactly two* sessions of a summer school. There are three seminars, each offered in four time slots (9–10, 10–11, 11–12, 12–13); every seminar is attended at most once and two sessions cannot occupy the same slot. The preference score of each session is:

|           | 9–10 | 10–11 | 11–12 | 12–13 |
|-----------|------|-------|-------|-------|
| Seminar 1 | 8    | 6     | 5     | 3     |
| Seminar 2 | 7    | 9     | 4     | 6     |
| Seminar 3 | 5    | 7     | 8     | 9     |

The student never wants *two consecutive hours* of class. The total preference is to be maximised.

**The model.**

The data are  $n = 3$  seminars,  $m = 4$  slots,  $q = 2$  sessions to attend and the score  $p_{sk}$ . With  $x_{sk} = 1$  if seminar  $s$  is attended in slot  $k$  ( $nm = 12$  binaries):

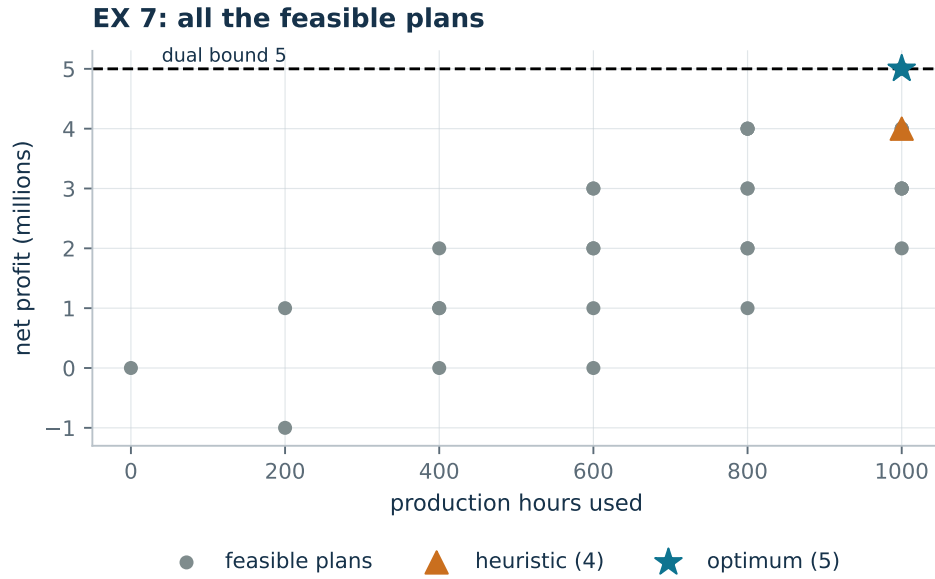


Figure 7: EX 7: every feasible plan in the hours–profit plane, with the heuristic, the optimum and the dual bound.

#### EX 8 on the instance — primal

$$\begin{aligned}
 & \max 8x_{11} + 6x_{12} + 5x_{13} + 3x_{14} + 7x_{21} + 9x_{22} + 4x_{23} + 6x_{24} + 5x_{31} + 7x_{32} + 8x_{33} + 9x_{34} \\
 \text{subject to } & x_{11} + x_{21} + x_{31} \leq 1 \\
 & x_{12} + x_{22} + x_{32} \leq 1 \\
 & x_{13} + x_{23} + x_{33} \leq 1 \\
 & x_{14} + x_{24} + x_{34} \leq 1 \\
 & x_{11} + x_{12} + x_{13} + x_{14} \leq 1 \\
 & x_{21} + x_{22} + x_{23} + x_{24} \leq 1 \\
 & x_{31} + x_{32} + x_{33} + x_{34} \leq 1 \\
 & x_{11} + x_{12} + x_{13} + x_{14} + x_{21} + x_{22} + x_{23} + x_{24} + x_{31} + x_{32} + x_{33} + x_{34} = 2 \\
 & x_{11} + x_{12} + x_{21} + x_{22} + x_{31} + x_{32} \leq 1 \\
 & x_{12} + x_{13} + x_{22} + x_{23} + x_{32} + x_{33} \leq 1 \\
 & x_{13} + x_{14} + x_{23} + x_{24} + x_{33} + x_{34} \leq 1 \\
 & x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34} \in \{0, 1\}
 \end{aligned}$$

**Description.** with  $q = 2$ : four slot constraints, three seminar constraints, one *exact* cardinality constraint and three non-adjacency constraints between consecutive slots.

#### Heuristic.

Constructive heuristic: take the feasible session with the highest score, then the best compatible one. Step 1: seminar 3 in slot 4 (score 9). Step 2: among the sessions that do not use seminar 3, do not occupy slot 4 and are not adjacent to it, the best is seminar 2 in slot 2 (score 9). In a **maximisation** the value is a *lower* bound:  $LB = 18$ .

### Dual and upper bound.

With  $\alpha_k \geq 0$  for the slots,  $\beta_s \geq 0$  for the seminars,  $\gamma$  free for the exact cardinality and  $\delta_k \geq 0$  for non-adjacency:

#### EX 8 on the instance — dual

$$\begin{array}{llllll}
 \min & \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \beta_1 + \beta_2 + \beta_3 + 2\gamma + \delta_1 + \delta_2 + \delta_3 \\
 \text{subject to} & & +\beta_1 & & +\gamma + \delta_1 & \geq 8 \\
 & \alpha_2 & & +\beta_1 & & +\gamma + \delta_1 + \delta_2 \geq 6 \\
 & & \alpha_3 & & +\beta_1 & +\gamma & +\delta_2 + \delta_3 \geq 5 \\
 & & & \alpha_4 + \beta_1 & & +\gamma & +\delta_3 \geq 3 \\
 \alpha_1 & & & & +\beta_2 & +\gamma + \delta_1 & \geq 7 \\
 & \alpha_2 & & & +\beta_2 & +\gamma + \delta_1 + \delta_2 \geq 9 \\
 & & \alpha_3 & & +\beta_2 & +\gamma & +\delta_2 + \delta_3 \geq 4 \\
 & & & \alpha_4 & +\beta_2 & +\gamma & +\delta_3 \geq 6 \\
 \alpha_1 & & & & & +\beta_3 & +\gamma + \delta_1 \geq 5 \\
 & \alpha_2 & & & & +\beta_3 & +\gamma + \delta_1 + \delta_2 \geq 7 \\
 & & \alpha_3 & & & +\beta_3 & +\gamma & +\delta_2 + \delta_3 \geq 8 \\
 & & & \alpha_4 & & +\beta_3 & +\gamma & +\delta_3 \geq 9 \\
 \alpha_1, \alpha_2, \alpha_3, \alpha_4 & & & & & & & \geq 0 \\
 & & & & \beta_1, \beta_2, \beta_3 & & & \geq 0 \\
 & & & & & & \gamma & \leq 0 \\
 & & & & & & & \delta_1, \delta_2, \delta_3 \geq 0
 \end{array}$$

**Description.** The variable  $\alpha_k$  is the price of time slot  $k$ ,  $\beta_s$  that of seminar  $s$ ,  $\gamma$  the price of the cardinality constraint and  $\delta_k$  that of the non-adjacency constraint between slots  $k$  and  $k+1$ . The objective adds up the right-hand sides of those constraints at those prices. The dual constraints are the columns of the  $x_{sk}$ : attending seminar  $s$  in slot  $k$  takes the slot, takes the seminar, consumes one unit of cardinality and touches the non-adjacency constraints containing  $k$ ; the total must cover the preference  $p_{sk}$ .

The shortest recipe: everything at zero except  $\bar{\gamma} = \max_{s,k} p_{sk} = 9$ . Every dual constraint becomes  $9 \geq p_{sk}$ , true by construction, and the value is  $q\bar{\gamma} = 2 \cdot 9 = 18$ . It reads: "two sessions are attended, and none is worth more than the best one".

| Dual recipe                                                                     | UB |
|---------------------------------------------------------------------------------|----|
| everything at zero except $\gamma = \max_{s,k} p_{sk}$                          | 18 |
| $\gamma = 0, \alpha_k = \max_s p_{sk}$ ("every slot is worth at most its best") | 34 |

The first is optimal for the relaxation; the second, apparently more natural, is almost twice as large. The recipe matters.

### The bounds.

The optimum is  $z(MILP) = 18$ : seminar 2 in slot 2 and seminar 3 in slot 4 — with a break in between, as required.

| $LB$ (constructive heuristic) | $z(MILP)$ | $z(LP)$ | $UB$ (dual by hand) | heuristic gap |
|-------------------------------|-----------|---------|---------------------|---------------|
| 18                            | 18        | 18      | 18                  | 0.0%          |

Here the two hand-built bounds *close* the problem: the heuristic is worth 18, the dual 18, so  $z(MILP) = 18$  is proved without the solver.

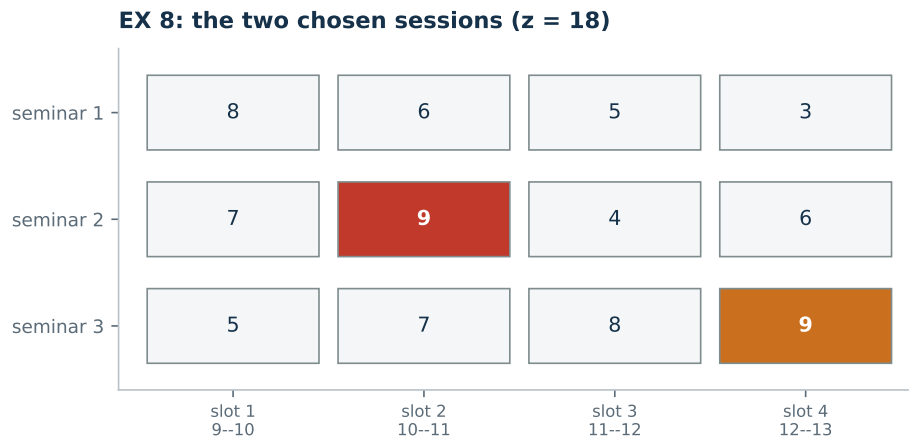


Figure 8: EX 8: the grid of preferences; in teal the two chosen sessions.

## EX 9 — Queens on the chessboard

Script: `python/ex09_queens.py`

[Notebook in Colab](#)

### EX 9 — the largest number of queens

On a  $4 \times 4$  chessboard one wants to place the largest number of queens so that no queen can capture another in a single move. In chess a queen moves any number of squares along the same row, the same column or either diagonal.

### The variables

One binary variable per square of the board.

$$x_{ij} = \begin{cases} 1 & \text{if there is a queen on square } (i, j), \\ 0 & \text{otherwise,} \end{cases} \quad \forall i \in \{1, 2, 3, 4\}, \forall j \in \{1, 2, 3, 4\}.$$

### The model on the instance

Sixteen columns, one per square of the board.

## EX 9 on the instance — primal

$$\begin{aligned}
& \max x_{11}+x_{12}+x_{13}+x_{14}+x_{21}+x_{22}+x_{23}+x_{24}+x_{31}+x_{32}+x_{33}+x_{34}+x_{41}+x_{42}+x_{43}+x_{44} \\
& \text{subject to } x_{11}+x_{12}+x_{13}+x_{14} \leq 1 \\
& \qquad \qquad \qquad x_{21}+x_{22}+x_{23}+x_{24} \leq 1 \\
& \qquad \qquad \qquad \qquad \qquad x_{31}+x_{32}+x_{33}+x_{34} \leq 1 \\
& \qquad \qquad \qquad \qquad \qquad \qquad \qquad x_{41}+x_{42}+x_{43}+x_{44} \leq 1 \\
& \qquad x_{11} \qquad \qquad +x_{21} \qquad \qquad +x_{31} \qquad \qquad +x_{41} \leq 1 \\
& \qquad \qquad x_{12} \qquad \qquad +x_{22} \qquad \qquad +x_{32} \qquad \qquad +x_{42} \leq 1 \\
& \qquad \qquad \qquad x_{13} \qquad \qquad +x_{23} \qquad \qquad +x_{33} \qquad \qquad +x_{43} \leq 1 \\
& \qquad \qquad \qquad \qquad x_{14} \qquad \qquad +x_{24} \qquad \qquad +x_{34} \qquad \qquad +x_{44} \leq 1 \\
& \qquad \qquad \qquad \qquad \qquad x_{13} \qquad \qquad +x_{24} \leq 1 \\
& \qquad \qquad x_{12} \qquad \qquad +x_{23} \qquad \qquad +x_{34} \leq 1 \\
& \qquad x_{11} \qquad \qquad +x_{22} \qquad \qquad +x_{33} \qquad \qquad +x_{44} \leq 1 \\
& \qquad \qquad \qquad x_{21} \qquad \qquad +x_{32} \qquad \qquad +x_{43} \leq 1 \\
& \qquad \qquad \qquad \qquad \qquad x_{31} \qquad \qquad +x_{42} \leq 1 \\
& \qquad \qquad x_{12} \qquad +x_{21} \leq 1 \\
& \qquad \qquad \qquad x_{13} \qquad +x_{22} \qquad +x_{31} \leq 1 \\
& \qquad \qquad \qquad \qquad x_{14} \qquad +x_{23} \qquad +x_{32} \qquad +x_{41} \leq 1 \\
& \qquad \qquad \qquad \qquad \qquad \qquad x_{24} \qquad +x_{33} \qquad +x_{42} \leq 1 \\
& \qquad \qquad \qquad \qquad \qquad \qquad \qquad \qquad x_{34} \qquad +x_{43} \leq 1 \\
& x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}, x_{31}, x_{32}, x_{33}, x_{34}, x_{41}, x_{42}, x_{43}, x_{44} \in \{0, 1\}
\end{aligned}$$

**Description.** A set packing on four families of lines — the 4 rows, the 4 columns and the diagonals in both directions — 18 constraints in total. The indexing trick is all here: the "descending" diagonals are the sets with  $i - j$  constant, the "ascending" ones the sets with  $i + j$  constant.

Each of the four corners sits on a diagonal that touches nothing else. Writing its constraint would give  $x_{ij} \leq 1$  on an already binary variable: a row that forbids nothing. What remains is 5 + 5 diagonals with at least two squares, and the model loses ten rows without losing a single solution.

### Heuristic: the primal bound.

Constructive heuristic row by row: in row  $i$  one picks the first free column that is not attacked by the queens already placed; if there is none, the row stays empty. It never backtracks. On the instance the constructive heuristic places queens on (1, 1) and (2, 3); row 3 has no free square left and is skipped; in row 4 it takes (4, 2).

$$z(MILP) \geq LB = 3.$$

### Dual: the dual bound.

The dual is a set covering: a non-negative variable per line —  $\alpha_i$  for row  $i$ ,  $\beta_j$  for column  $j$ ,  $\gamma_k$  and  $\delta_k$  for the two families of diagonals — and every square must be covered by at least one of the four lines crossing it.

#### EX 9 on the instance — dual

$$\begin{aligned}
 & \min \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \beta_1 + \beta_2 + \beta_3 + \beta_4 + \gamma_{-1} + \gamma_0 + \gamma_1 + \gamma_2 + \gamma_3 + \delta_2 + \delta_3 + \delta_4 + \delta_5 + \delta_6 \\
 & \text{subject to} \\
 & \alpha_1 + \beta_1 + \gamma_1 \geq 1 \\
 & \alpha_1 + \beta_2 + \gamma_0 + \delta_2 \geq 1 \\
 & \alpha_1 + \beta_3 + \gamma_{-1} + \delta_3 \geq 1 \\
 & \alpha_1 + \beta_4 + \delta_4 \geq 1 \\
 & \alpha_2 + \beta_1 + \gamma_2 + \delta_2 \geq 1 \\
 & \alpha_2 + \beta_2 + \gamma_1 + \delta_3 \geq 1 \\
 & \alpha_2 + \beta_3 + \gamma_0 + \delta_4 \geq 1 \\
 & \alpha_2 + \beta_4 + \gamma_{-1} + \delta_5 \geq 1 \\
 & \alpha_3 + \beta_1 + \gamma_3 + \delta_3 \geq 1 \\
 & \alpha_3 + \beta_2 + \gamma_2 + \delta_4 \geq 1 \\
 & \alpha_3 + \beta_3 + \gamma_1 + \delta_5 \geq 1 \\
 & \alpha_3 + \beta_4 + \gamma_0 + \delta_6 \geq 1 \\
 & \alpha_4 + \beta_1 + \delta_4 \geq 1 \\
 & \alpha_4 + \beta_2 + \gamma_3 + \delta_5 \geq 1 \\
 & \alpha_4 + \beta_3 + \gamma_2 + \delta_6 \geq 1 \\
 & \alpha_4 + \beta_4 + \gamma_1 \geq 1 \\
 & \alpha_1, \alpha_2, \alpha_3, \alpha_4 \geq 0 \\
 & \beta_1, \beta_2, \beta_3, \beta_4 \geq 0 \\
 & \gamma_{-1}, \gamma_0, \gamma_1, \gamma_2, \gamma_3 \geq 0 \\
 & \delta_2, \delta_3, \delta_4, \delta_5, \delta_6 \geq 0
 \end{aligned}$$

For example one can use a single line:  $\bar{\alpha}_i = 1$  for every row, everything else zero. Every square  $(i, j)$  has  $\alpha_i = 1 \geq 1$ , so the solution is feasible and is worth 4. It is the dual translation of the sentence "at most one queen sits in each row".

$$z(MILP) \leq UB = 4.$$

### Results.

|                   | LB | UB | $z(LP)$ | $z(LP^+)$ | $z(MILP)$ |
|-------------------|----|----|---------|-----------|-----------|
| EX 9 queens (max) | 3  | 4  | 4       | 4         | 4         |

The dual bound is attained: the solution found is optimal, and one knows it without trusting the solver. The constructive heuristic, on the other hand, stops at 3: it is the heuristic, not the bound, that leaves the 25% gap.

### Two variants.

On  $n \times n$  boards with  $n = 5$  and  $n = 6$  the optimum is always  $n$ , and the certificate  $\bar{\alpha}_i = 1$  proves it. For  $n = 2$  and  $n = 3$ , instead, the optimum is 1 and 2: the bound  $n$  stays valid but is not attainable, and it is the first case in which this dual recipe does not settle the problem. The  $4 \times 4$  board of the exercise is the smallest on which  $n$  queens actually fit, and therefore the smallest on which the certificate settles it. Removing the diagonal constraints one gets *rooks* instead of queens: the model becomes an assignment, the matrix is totally unimodular and the LP relaxation already gives an integer value.

### EX 9: constructive heuristic against optimum

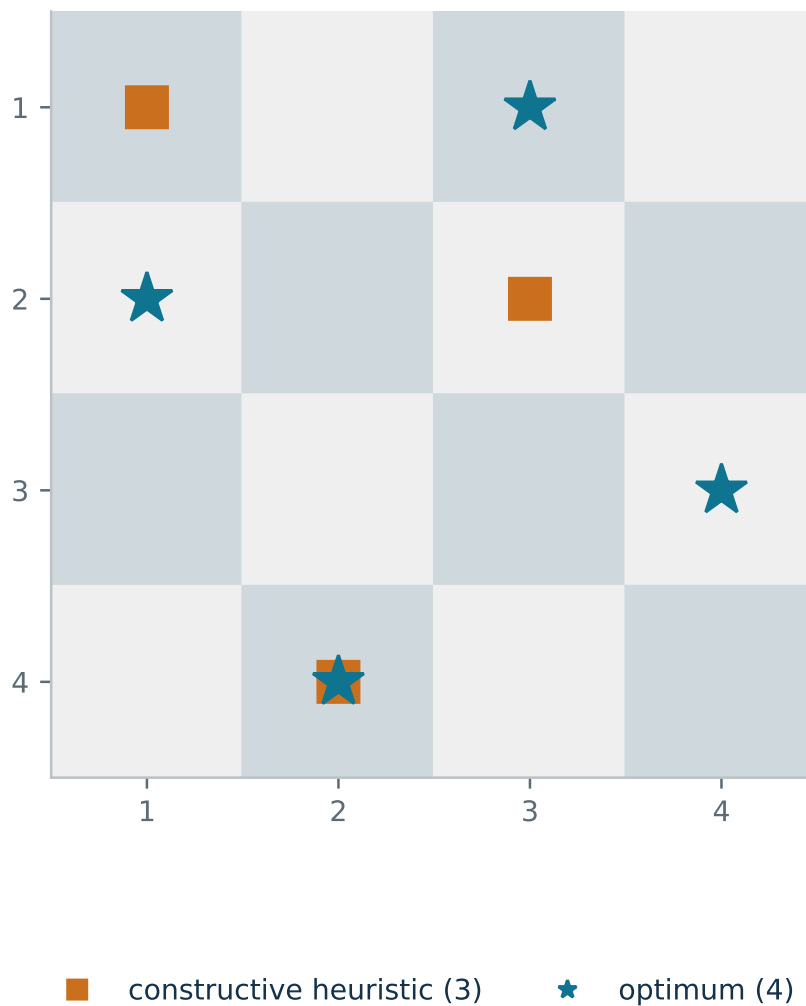


Figure 9: EX 9: the three queens placed by the heuristic and the four of the optimal solution.

### EX 10 — Tools of a CNC machine

Script: `python/ex10_tools.py`

[Notebook in Colab](#)

**EX 10 — selection with a tool set**

A CNC machine can perform five operations; each one requires a subset of six tools and, if performed, yields a profit:

| Operation | Required tools | Profit (€) |
|-----------|----------------|------------|
| 1         | 1, 3, 4, 5     | 2000       |
| 2         | 1, 2, 6        | 1500       |
| 3         | 1, 2, 3, 5     | 1800       |
| 4         | 2, 4, 5        | 1700       |
| 5         | 5, 6           | 800        |

The magazine of the machine holds at most  $k = 4$  tools at a time. The configuration of maximum profit is wanted.

**The model.**

The data are  $n = 5$  operations of profit  $p_i$ ,  $m = 6$  tools and a magazine holding  $k = 4$  tools;  $T_i$  is the set of tools required by operation  $i$ . With  $x_i = 1$  if operation  $i$  is performed ( $n$  binaries) and  $y_j = 1$  if tool  $j$  is loaded ( $m$  binaries):

**EX 10 on the instance — primal**

$$\begin{aligned}
 & \max 2000x_1 + 1500x_2 + 1800x_3 + 1700x_4 + 800x_5 \\
 & \text{subject to} \\
 & \quad y_1 + y_2 + y_3 + y_4 + y_5 + y_6 \leq 4 \\
 & \quad x_1 - y_1 \leq 0 \\
 & \quad x_1 - y_3 \leq 0 \\
 & \quad x_1 - y_4 \leq 0 \\
 & \quad x_1 - y_5 \leq 0 \\
 & \quad x_2 - y_1 \leq 0 \\
 & \quad x_2 - y_2 \leq 0 \\
 & \quad x_2 - y_6 \leq 0 \\
 & \quad x_3 - y_1 \leq 0 \\
 & \quad x_3 - y_2 \leq 0 \\
 & \quad x_3 - y_3 \leq 0 \\
 & \quad x_3 - y_5 \leq 0 \\
 & \quad x_4 - y_2 \leq 0 \\
 & \quad x_4 - y_4 \leq 0 \\
 & \quad x_4 - y_5 \leq 0 \\
 & \quad x_5 - y_5 \leq 0 \\
 & \quad x_5 - y_6 \leq 0 \\
 & \quad x_1, x_2, x_3, x_4, x_5 \in \{0, 1\} \\
 & \quad y_1, y_2, y_3, y_4, y_5, y_6 \in \{0, 1\}
 \end{aligned}$$

**Description.** One magazine constraint and  $\sum_i |T_i| = 4 + 3 + 4 + 3 + 2 = 16$  link constraints. It is the *disaggregated* activation of Section “Activation: aggregated and disaggregated form”

read the other way round: here the operation is the object and the tools are the resources, and the operation needs *all* of its tools — a conjunction in the consequent, which by the table of Chapter “Logic and binary variables” costs one constraint per tool.

### Heuristic.

Constructive heuristic by decreasing profit: the operations are scanned from the most profitable and the tool set of each is loaded, if the magazine allows it.

- **Operation 1** (2000, tools 1,3,4,5): 4 new tools are needed and the magazine reaches exactly 4: it runs.
- **Operations 3, 4, 2, 5**: each would need at least one more tool, and the magazine is full: they are discarded.

In a **maximisation** the value is a *lower* bound:  $LB = 2000$ .

### Dual and upper bound.

Relaxing to  $x_i, y_j \geq 0$ , with  $\alpha \geq 0$  for the magazine constraint and  $\beta_{ij} \geq 0$  for each link constraint:

#### EX 10 on the instance — dual

$$\begin{array}{llllll}
 \min & 4\alpha & & & & \\
 \text{subject to} & \beta_{11} + \beta_{13} + \beta_{14} + \beta_{15} & & & & \geq 2000 \\
 & & \beta_{21} + \beta_{22} + \beta_{26} & & & \geq 1500 \\
 & & & \beta_{31} + \beta_{32} + \beta_{33} + \beta_{35} & & \geq 1800 \\
 & & & & \beta_{42} + \beta_{44} + \beta_{45} & \geq 1700 \\
 & & & & & \beta_{55} + \beta_{56} \geq 800 \\
 & -\alpha + \beta_{11} & +\beta_{21} & +\beta_{31} & & \leq 0 \\
 & -\alpha & & +\beta_{22} & +\beta_{32} & +\beta_{42} \leq 0 \\
 & -\alpha & +\beta_{13} & & +\beta_{33} & \leq 0 \\
 & -\alpha & & +\beta_{14} & & +\beta_{44} \leq 0 \\
 & -\alpha & & +\beta_{15} & & +\beta_{35} + \beta_{45} + \beta_{55} \leq 0 \\
 & -\alpha & & & +\beta_{26} & +\beta_{56} \leq 0 \\
 & \alpha & & & & \geq 0 \\
 & \beta_{11}, \beta_{13}, \beta_{14}, \beta_{15}, \beta_{21}, \beta_{22}, \beta_{26}, \beta_{31}, \beta_{32}, \beta_{33}, \beta_{35}, \beta_{42}, \beta_{44}, \beta_{45}, \beta_{55}, \beta_{56} & \geq 0
 \end{array}$$

**Description.** It reads like this:  $\beta_{ij}$  is the share of the profit of operation  $i$  attributed to tool  $j$ ; the first group asks the shares to cover the whole profit, the second that no tool receives more than  $\alpha$ ; the objective pays  $\alpha$  for each of the  $k$  slots in the magazine.

**The recipe.** The profit of each operation is spread in equal parts over the tools it needs,  $\bar{\beta}_{ij} = p_i/|T_i|$ : the first group is satisfied with equality by construction. Then  $\bar{\alpha} = \max_j \sum_{i:j \in T_i} \bar{\beta}_{ij}$ , the load of the most demanded tool.

| $\bar{\beta}_{ij} = p_i/ T_i $ | load of the tool |        |     |        |               |     |
|--------------------------------|------------------|--------|-----|--------|---------------|-----|
|                                | 1                | 2      | 3   | 4      | 5             | 6   |
| op. 1: $2000/4 = 500$          | 500              |        | 500 | 500    | 500           |     |
| op. 2: $1500/3 = 500$          | 500              | 500    |     |        |               | 500 |
| op. 3: $1800/4 = 450$          | 450              | 450    | 450 |        | 450           |     |
| op. 4: $1700/3 = 1700/3$       |                  | 1700/3 |     | 1700/3 | 1700/3        |     |
| op. 5: $800/2 = 400$           |                  |        |     |        | 400           | 400 |
| total                          | 1450             | 4550/3 | 950 | 3200/3 | <b>5750/3</b> | 900 |

The critical tool is number 5, required by four operations out of five: hence  $\bar{\alpha} = 5750/3$  and

$$UB = k \bar{\alpha} = 4 \cdot \frac{5750}{3} = \frac{23000}{3} \approx 7666.7.$$

A choice that comes naturally is  $\bar{\alpha} = 2000$  with all  $\bar{\beta}_{ij} = 900$ : the constraints on the operations are satisfied ( $2 \cdot 900 = 1800 \geq 800$  is the tightest case) and the value would be  $4 \cdot 2000 = 8000$ . **But that solution is not feasible.** The second group of constraints asks  $\sum_{i:j \in T_i} \beta_{ij} \leq \alpha$ , and tool 5 serves *four* operations: it would receive  $4 \cdot 900 = 3600 > 2000$ . The violation is 1600. The number 8000 is not a bound: it does not come from any feasible dual solution.

This is why the course protocol does not ask to *propose* a dual solution but to *check* it: the script verifies every constraint, and an **assert** fails if the violation exceeds the tolerance. With the correct recipe the bound is  $23000/3 \approx 7666.7$ , that is, *better* than the one hoped for by getting it wrong.

### The bounds.

$z(MILP) = 2500$ : tools 2, 4, 5, 6 are loaded and operations 4 (1700) and 5 (800) are performed. No four-tool configuration yields more: with  $\{1, 3, 4, 5\}$  only operation 1 runs (2000), with  $\{1, 2, 3, 5\}$  only operation 3 (1800).

| $LB$ (constructive heuristic) | $z(MILP)$ | $z(LP)$ | $z(LP^+)$ | $UB$ (dual by hand) | heuristic gap |
|-------------------------------|-----------|---------|-----------|---------------------|---------------|
| 2000                          | 2500      | 5200    | 5200      | 23000/3             | 20.0%         |

### Why the relaxation is so weak here

$z(LP) = 5200$  against  $z(MILP) = 2500$ : more than double. In the continuous problem one can load "a bit" of every tool — for instance  $y_j = 2/3$  for six tools, which respects  $\sum_j y_j \leq 4$  — and run a fraction of *every* operation. The integrality of the magazine is everything: this is the case where hand-built bounds, however well constructed, are not enough and the work has to be done by the solver. One way to tighten is to add the valid inequalities  $\sum_{j \in T_i} y_j \geq |T_i| x_i$  (the aggregated form of the links), or to observe that any operation with  $|T_i| > k$  can be fixed to zero.

## EX 11 — Shoes with a minimum production threshold

Script: `python/ex11_shoes_threshold.py`

[Notebook in Colab](#)

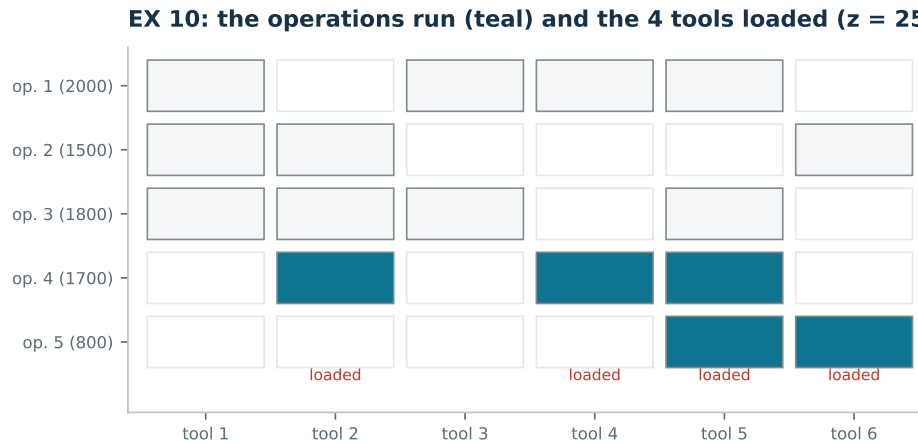


Figure 10: EX 10: the operation-tool matrix. In teal the operations performed; the annotated columns are the four loaded tools.

### EX 11 — three types of shoe with a threshold

A shoe company produces hiking boots, loafers and walking shoes. The table reports the resources needed for one pair, the minimum quantity to be produced if the type is produced, and the selling price:

| Type          | leather (g) | hours | nails | minimum | price |
|---------------|-------------|-------|-------|---------|-------|
| hiking boots  | 850         | 3     | 20    | 100     | 150   |
| loafers       | 600         | 2     | 15    | 200     | 120   |
| walking shoes | 700         | 2.5   | 20    | 150     | 130   |

There are 120 000 grams of leather, 7000 machine hours and 40 000 nails available. The goal is to maximise the revenue.

### The variables

Two families of variables, one per type of shoe:  $x_j$ , the pairs produced of type  $j$ , integer and non-negative, and  $y_j$ , worth 1 if type  $j$  is produced and 0 otherwise.

### The model.

It is again model 9.3 without the bonus, with three resources. The big-Ms read off the data are  $M = (141, 200, 171)$ , all limited by the leather. For the loafers threshold and maximum coincide (200): the variable  $x_2$  is in fact a binary multiplied by 200.

## EX 11 on the instance — primal

$$\begin{array}{rcll}
\max & 150x_1 + 120x_2 + 130x_3 & & \\
\text{subject to} & 850x_1 + 600x_2 + 700x_3 & & \leq 120000 \\
& 3x_1 + 2x_2 + \frac{5}{2}x_3 & & \leq 7000 \\
& 20x_1 + 15x_2 + 20x_3 & & \leq 40000 \\
& x_1 & -100y_1 & \geq 0 \\
& & x_2 & -200y_2 \geq 0 \\
& & & x_3 & -150y_3 \geq 0 \\
& x_1 & & -141y_1 & \leq 0 \\
& & x_2 & & -200y_2 \leq 0 \\
& & & x_3 & -171y_3 \leq 0 \\
& x_1, & x_2, & x_3 & \in \mathbb{Z}_{\geq 0} \\
& & & & y_1, & y_2, & y_3 \in \{0, 1\}
\end{array}$$

**Description.** The first three rows are the resources — leather, hours and nails — and each one forbids using more than there is. The next three are the *minimum thresholds*: if a type is produced, at least 100, 200 or 150 pairs of it are. The three after that are the *activations* with the big-M read off the data: with  $y_j = 0$  production is pinned to zero. The last two rows declare the productions integer and the choices binary.

**Heuristic: the primal bound.**

Constructive heuristic on the price per gram of leather:  $3/17$  for the boots,  $1/5$  for the loafers,  $13/70$  for the walking shoes. One starts from the loafers, produces 200 of them and the leather runs out exactly. No other type reaches its own threshold.

$$z(MILP) \geq LB = 24\,000.$$

**Dual: the dual bound.**

With  $\ell = \beta = 0$  and a single resource priced:

$$\text{leather: } \frac{1}{5} \Rightarrow 24\,000, \quad \text{hours: } 60 \Rightarrow 420\,000, \quad \text{nails: } 8 \Rightarrow 320\,000.$$

The better bound is the one from the leather:

$$z(MILP) \leq UB = 24\,000.$$

**Results.**

|                                  | $LB$   | $UB$   | $z(LP)$ | $z(LP^+)$ | $z(MILP)$ |
|----------------------------------|--------|--------|---------|-----------|-----------|
| EX 11 shoes with threshold (max) | 24 000 | 24 000 | 24 000  | 24 000    | 24 000    |

All the numbers coincide: the sandwich closes immediately. The reason is that a single resource is tight, the leather, and the type that gets the most out of it per gram uses it up on its own, staying within its own threshold and its own maximum. It is a didactically useful case precisely

because it is easy: it shows that when the dual bound matches a feasible solution, the problem is over.

Two variations change the picture. Doubling the thresholds, the one for the loafers becomes 400 pairs while the leather allows at most 200: no type reaches its own threshold any more and production stops altogether,  $z(MILP) = 0$ . Raising the leather to 200 000 grams, 333 loafers are produced for 39 960 euros, and the big-Ms must be recomputed: they become (235, 333, 285). Big-M values are not constants of the model, but functions of the data of the instance.

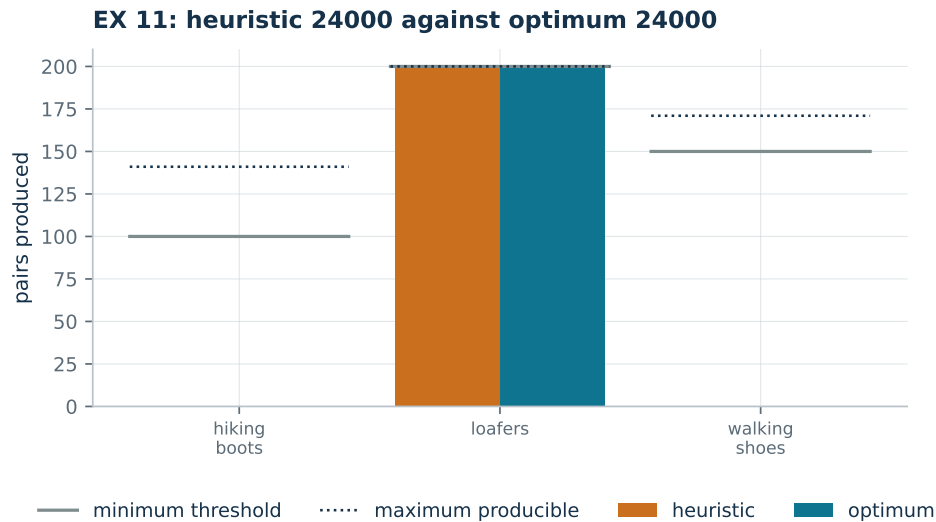


Figure 11: EX 11: the pairs produced by the heuristic and the optimal ones, with the minimum threshold and the maximum producible of each type.

## EX 12 — Balancing between two workers

Script: `python/ex12_balancing.py`

[Notebook in Colab](#)

### EX 12 — workloads as balanced as possible

Four indivisible jobs of duration 2, 3, 6 and 7 must be assigned to two workers so that their workloads are as balanced as possible.

"Balanced workloads" is not an objective: it is a word. The two linear translations are those of Section "Min-max, max-min and the range": minimise the *maximum* of the loads, or minimise their *difference*. Since the total  $q = 18$  is constant, the identity

$$\max(w_1, w_2) = \frac{q}{2} + \frac{|w_1 - w_2|}{2}$$

holds, so the two models have the *same* optimal solutions — but different optimal values: 9 for the first, 0 for the second. Reporting "9" as the "difference between the loads" is a mistake, and so is reporting "0" as the "maximum load".

### The model (min-max).

The data are  $n = 4$  jobs of duration  $d_j$  and the total load  $q = \sum_{j=1}^n d_j = 18$ . With  $x_j = 1$  if job  $j$  goes to worker 1 and  $z \geq 0$  the maximum load:

#### EX 12 on the instance — primal

$$\begin{aligned}
 & \min && z \\
 & \text{subject to} && 2x_1 + 3x_2 + 6x_3 + 7x_4 - z \leq 0 \\
 & && -2x_1 - 3x_2 - 6x_3 - 7x_4 - z \leq -18 \\
 & && x_1, \quad x_2, \quad x_3, \quad x_4 \in \{0, 1\} \\
 & && z \geq 0
 \end{aligned}$$

**Description.** The objective is the load of the busier worker. The first constraint says that  $z$  is no smaller than the load of worker 1; the second that it is no smaller than that of worker 2, which equals  $q - \sum_j d_j x_j$ . Together they impose  $z \geq \max(w_1, w_2)$ , and the minimisation turns the inequality into an equality on the busier of the two.

### Heuristic.

LPT (Chapter “Constructive heuristics”): the jobs in decreasing order of duration, each to the least loaded worker. Job 4 (7) to worker 1; job 3 (6) to worker 2; job 2 (3) to worker 2, which reaches 9; job 1 (2) to worker 1, which reaches 9. Loads (9, 9), so  $UB = 9$ .

### Dual and lower bound.

Relaxing to  $x_j \geq 0$  and using the conversion table — a *minimisation* primal with  $\leq$  constraints, hence dual variables  $\pi_1, \pi_2 \leq 0$ :

#### EX 12 on the instance — dual

$$\begin{aligned}
 & \max && -18\pi_2 \\
 & \text{subject to} && 2\pi_1 - 2\pi_2 \leq 0 \\
 & && 3\pi_1 - 3\pi_2 \leq 0 \\
 & && 6\pi_1 - 6\pi_2 \leq 0 \\
 & && 7\pi_1 - 7\pi_2 \leq 0 \\
 & && -\pi_1 - \pi_2 \leq 1 \\
 & && \pi_1, \quad \pi_2 \leq 0
 \end{aligned}$$

**Description.** The variables  $\pi_1$  and  $\pi_2$  are the (non-positive) prices of the two min-max constraints. The objective prices the total load  $q$  at  $\pi_2$ . The first group of constraints are the columns of the  $x_j$ : moving job  $j$  from one worker to the other shifts both loads by  $d_j$ , and the balance cannot be positive because the  $x_j$  do not appear in the objective. The last constraint is the column of  $z$ , which appears in both constraints with coefficient  $-1$  and in the primal objective with cost 1.

Since  $d_j > 0$ , the first constraints say  $\pi_1 \leq \pi_2$ . Setting  $\pi_1 = \pi_2 = t$ , the constraint from  $z$  gives  $-2t \leq 1$ , that is  $t \geq -1/2$ ; the objective  $-qt$  is largest at  $t = -1/2$  and equals  $LB = q/2 = 9$ . Meaning: "the two loads add up to  $q$ , so the larger one is at least  $q/2$ ".

### The same thing with non-negative dual variables

Many treatments prefer dual variables  $\geq 0$ . It suffices to set  $\alpha = -\pi_1$  and  $\beta = -\pi_2$ : the dual becomes  $\max q\beta$  with  $\alpha \leq \beta$  and  $\alpha + \beta \leq 1$ , and  $\alpha = \beta = 1/2$  gives 9 again. It is the *same* solution written with the other sign; what one cannot do is declare variables  $\geq 0$  and then use one of them negative.

### The bounds.

$z(MILP) = 9$  with jobs  $\{1, 4\}$  to the first worker and  $\{2, 3\}$  to the second: loads  $(9, 9)$ , perfectly balanced.

| $UB$ (LPT) | $LB$ (dual by hand) | $z(LP)$ | $z(MILP)$ | heuristic gap |
|------------|---------------------|---------|-----------|---------------|
| 9          | 9                   | 9       | 9         | 0.0%          |

All the numbers coincide: here the two hand-built bounds meet, and optimality is proved without solving the MILP. This happens because  $q$  is even and the jobs can be split exactly in half. With  $q$  odd the dual of the relaxation would still give  $q/2$ , which is not an integer: it is the integrality of the loads that allows strengthening it to  $\lceil q/2 \rceil$  — and that is a combinatorial argument, not a dual bound.

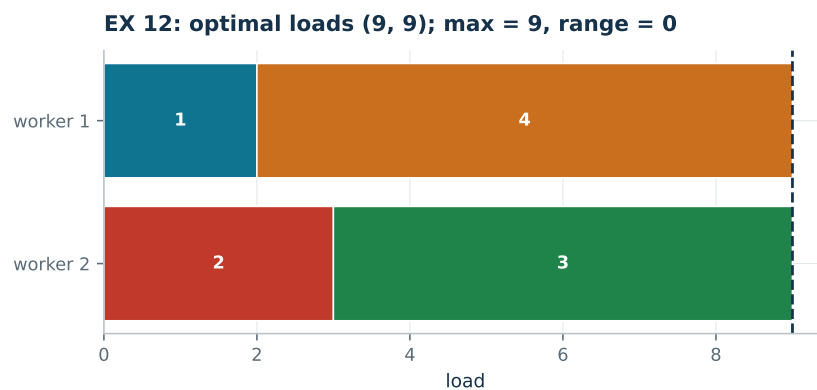


Figure 12: EX 12: the optimal loads. The dashed line is  $q/2$ , the dual bound.

## EX 13 — Emergency department shifts

Script: `python/ex13_shifts.py`

[Notebook in Colab](#)

### EX 13 — weekly shifts

In an emergency department the number of full-time staff members required is given for every day of the week:

| Day   | Mon | Tue | Wed | Thu | Fri | Sat | Sun |
|-------|-----|-----|-----|-----|-----|-----|-----|
| staff | 10  | 8   | 12  | 9   | 9   | 7   | 8   |

One full-time member can be replaced by two part-time members. The shift pattern is fixed: four consecutive full-time days, then one part-time day, then two rest days. The weekly cost of a worker depends on the days worked: 100 euros for every weekday (Monday to Friday),

110 for Saturday and 130 for Sunday; these costs are halved on part-time days. The goal is the weekly plan of minimum cost.

### The model.

There are  $n = 7$  shift patterns, one per starting day, and  $m = 7$  days to cover, with requirement  $b_i$ . With  $x_j$  the number of workers following pattern  $j$  (integer variables),  $c_j$  the weekly cost of pattern  $j$  and  $a_{ij}$  the share of day  $i$  covered by pattern  $j$  (1 on the four full days,  $1/2$  on the part-time day, 0 on the two rest days):

#### EX 13 on the instance — primal

$$\begin{aligned}
 & \min 450x_1 + 455x_2 + 475x_3 + 490x_4 + 490x_5 + 490x_6 + 480x_7 \\
 & \text{subject to} \quad x_1 \quad \quad \quad + \frac{1}{2}x_4 \quad + x_5 \quad + x_6 \quad + x_7 \geq 10 \\
 & \quad \quad x_1 \quad + x_2 \quad \quad \quad + \frac{1}{2}x_5 \quad + x_6 \quad + x_7 \geq 8 \\
 & \quad \quad x_1 \quad + x_2 \quad + x_3 \quad \quad \quad + \frac{1}{2}x_6 \quad + x_7 \geq 12 \\
 & \quad \quad x_1 \quad + x_2 \quad + x_3 \quad + x_4 \quad \quad \quad + \frac{1}{2}x_7 \geq 9 \\
 & \quad \quad \frac{1}{2}x_1 \quad + x_2 \quad + x_3 \quad + x_4 \quad + x_5 \quad \quad \quad \geq 9 \\
 & \quad \quad \quad \frac{1}{2}x_2 \quad + x_3 \quad + x_4 \quad + x_5 \quad + x_6 \quad \quad \quad \geq 7 \\
 & \quad \quad \quad \quad \frac{1}{2}x_3 \quad + x_4 \quad + x_5 \quad + x_6 \quad + x_7 \geq 8 \\
 & \quad \quad x_1, \quad x_2, \quad x_3, \quad x_4, \quad x_5, \quad x_6, \quad x_7 \in \mathbb{Z}_{\geq 0}
 \end{aligned}$$

**Description.** It is a *set covering* with coefficients 1 and  $1/2$  and integer, non-binary variables. The costs are computed from the data:

| Pattern        | full days          | part-time | cost |
|----------------|--------------------|-----------|------|
| 1 (starts Mon) | Mon, Tue, Wed, Thu | Fri       | 450  |
| 2 (starts Tue) | Tue, Wed, Thu, Fri | Sat       | 455  |
| 3 (starts Wed) | Wed, Thu, Fri, Sat | Sun       | 475  |
| 4 (starts Thu) | Thu, Fri, Sat, Sun | Mon       | 490  |
| 5 (starts Fri) | Fri, Sat, Sun, Mon | Tue       | 490  |
| 6 (starts Sat) | Sat, Sun, Mon, Tue | Wed       | 490  |
| 7 (starts Sun) | Sun, Mon, Tue, Wed | Thu       | 480  |

For instance pattern 3 costs  $100 + 100 + 100 + 110$  for the four full days (Wed, Thu, Fri, Sat) plus  $130/2 = 65$  for the part-time Sunday, that is 475 in total.

### Heuristic: the primal bound.

Constructive covering heuristic: while some demand is uncovered one adds a copy of the pattern with the lowest ratio

$$\frac{\text{cost}}{\text{demand actually covered}}.$$

The ratio is recomputed at every step, because the residual demand changes. On the instance the constructive heuristic adds 18 shifts: eight of pattern 1 (at 100 euros per unit of demand covered), then four of pattern 3, two of pattern 5, one of pattern 4 and three of pattern 7, with steadily worsening ratios, up to 480 for the last unit.

$$z(\text{MILP}) \leq UB = 8410.$$

### Dual: the dual bound.

The dual is a variable  $\pi_i \geq 0$  for each day, the price attached to covering it, and every shift pattern must cost at least as much as the coverage it offers.

#### EX 13 on the instance — dual

$$\begin{aligned}
 & \max 10\pi_1 + 8\pi_2 + 12\pi_3 + 9\pi_4 + 9\pi_5 + 7\pi_6 + 8\pi_7 \\
 & \text{subject to} \quad \pi_1 + \pi_2 + \pi_3 + \pi_4 + \frac{1}{2}\pi_5 \leq 450 \\
 & \quad \quad \quad \pi_2 + \pi_3 + \pi_4 + \pi_5 + \frac{1}{2}\pi_6 \leq 455 \\
 & \quad \quad \quad \pi_3 + \pi_4 + \pi_5 + \pi_6 + \frac{1}{2}\pi_7 \leq 475 \\
 & \quad \quad \quad \frac{1}{2}\pi_1 + \pi_4 + \pi_5 + \pi_6 + \pi_7 \leq 490 \\
 & \quad \quad \quad \pi_1 + \frac{1}{2}\pi_2 + \pi_5 + \pi_6 + \pi_7 \leq 490 \\
 & \quad \quad \quad \pi_1 + \pi_2 + \frac{1}{2}\pi_3 + \pi_6 + \pi_7 \leq 490 \\
 & \quad \quad \quad \pi_1 + \pi_2 + \pi_3 + \frac{1}{2}\pi_4 + \pi_7 \leq 480 \\
 & \quad \quad \quad \pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6, \pi_7 \geq 0
 \end{aligned}$$

For example one can use the *best ratio* one: put the same price  $t$  on all the days. Every pattern covers  $4 + 1/2 = 9/2$  person-days, so the dual constraint becomes  $\frac{9}{2}t \leq c_j$  for every pattern, and the largest feasible value is

$$t = \min_j \frac{c_j}{9/2} = \min\left(100, \frac{910}{9}, \frac{950}{9}, \frac{980}{9}, \frac{980}{9}, \frac{980}{9}, \frac{320}{3}\right) = 100,$$

attained by pattern 1. The total demand is  $10 + 8 + 12 + 9 + 9 + 7 + 8 = 63$ , so

$$z(MILP) \geq LB = 100 \cdot 63 = 6300.$$

### Results.

|              | UB    | LB    | $z(LP)$   | $z(LP^+)$ | $z(MILP)$ |
|--------------|-------|-------|-----------|-----------|-----------|
| EX 13 shifts | 8 410 | 6 300 | 115970/17 | 115970/17 | 7 060     |

The optimal plan hires 5 workers on pattern 1, 4 on pattern 3, 1 on pattern 4, 2 on pattern 5 and 3 on pattern 7, at a cost of 7060. The coverage is saturated on Wednesday, Saturday and Sunday, and in surplus on the other days. The heuristic gap is 19.1%; the difference  $z(MILP) - z(LP) = 7060 - 115970/17 \approx 238$  is the price of integrality, that is of the fact that people are hired one at a time.

### Is the part-time day worth it?

Let us compare the contract of the statement with an alternative one: four full days and *three* rest days, with no part-time. The minimum price per day covered is the same in both cases, 100 euros. And yet the optimum of the second contract is 6710, against the 7060 of the first: the part-time day is worth less. The reason is that the half day falls on a fixed day and often ends up where the coverage is already there; the flexibility lost costs more than the extra coverage is worth. It is an example in which the conclusion "more coverage at the same unit price is always better" is false, and only the model shows it.

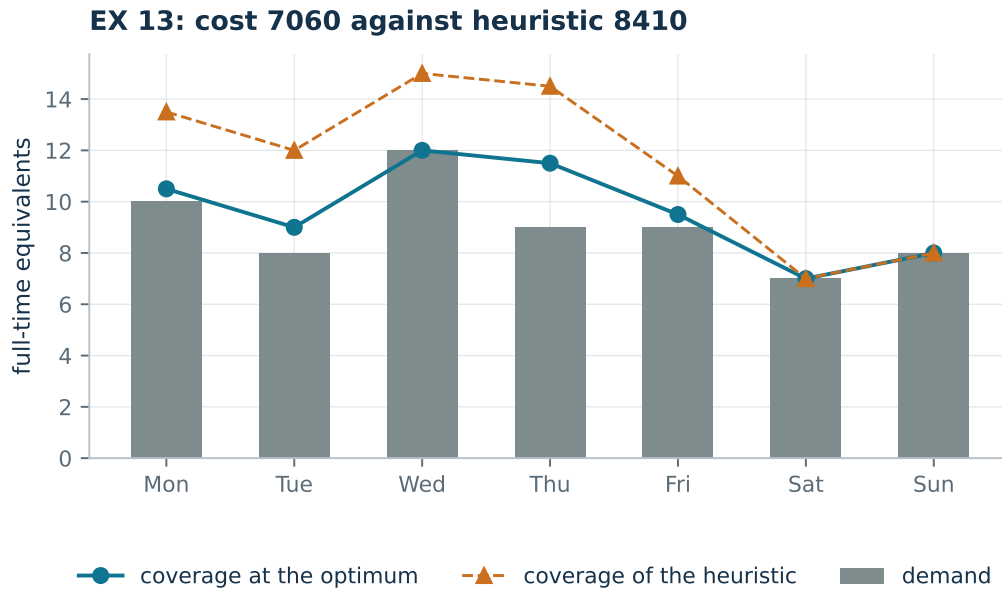


Figure 13: EX 13: daily demand and coverage, in the heuristic solution and in the optimal one.

## EX 14 — Shoes: production, inventory and hirings

Script: `python/ex14_shoes.py`

[Notebook in Colab](#)

### EX 14 — production and workforce over three months

Over the next three months a shoe factory must meet monthly demands of 3000, 5000 and 7000 pairs of shoes. At the start of the first month it has 500 pairs in stock and employs 100 workers. Every worker is paid 1500 euros a month and works 160 hours; producing one pair requires 4 labour hours and 15 euros of raw materials. At the start of every month workers may be hired, at a cost of 100 euros each for training. Storing one unsold pair at the end of the month costs 3 euros. The factory wants the production and workforce plan of minimum total cost.

### The model.

It is model 9.2A with one more datum, the initial inventory. With  $x_t$  pairs produced,  $s_t$  inventory at the end of the month,  $y_t$  workers on duty and  $z_t$  workers hired:

## EX 14 on the instance

$$\begin{array}{llllllllll}
\min & 15x_1 + 15x_2 + 15x_3 + 3s_1 + 3s_2 + 1500y_1 + 1500y_2 + 1500y_3 + 100z_1 + 100z_2 + 100z_3 \\
\text{subject to} & x_1 & & & -s_1 & & & & & & = 2500 \\
& & x_2 & & +s_1 & -s_2 & & & & & = 5000 \\
& & & x_3 & & +s_2 & & & & & = 7000 \\
& -4x_1 & & & & & +160y_1 & & & & \geq 0 \\
& & -4x_2 & & & & +160y_2 & & & & \geq 0 \\
& & & -4x_3 & & & +160y_3 & & & & \geq 0 \\
& & & & y_1 & & & -z_1 & & & = 100 \\
& & & & -y_1 & +y_2 & & & -z_2 & & = 0 \\
& & & & & -y_2 & +y_3 & & & -z_3 & = 0 \\
& x_1, & x_2, & x_3 & & & & & & & \geq 0 \\
& & & & s_1, & s_2 & & & & & \geq 0 \\
& & & & & & y_1, & y_2, & y_3 & & \in \mathbb{Z}_{\geq 0} \\
& & & & & & & & & z_1, & z_2, & z_3 \in \mathbb{Z}_{\geq 0}
\end{array}$$

**Description.** The demand of the first month is  $3000 - 500 = 2500$  net pairs; 14 500 pairs must be produced in total. Here the workforce appears in two families of variables at once,  $y_t$  and  $z_t$ , tied by the staff balance: it is the mixed formulation one obtains by keeping both readings of problem 9.2.

### Heuristic: the primal bound.

"Just in time" production: every month the net demand is produced, with no inventory, hiring as many workers as needed. Month 1: 10 000 hours, 63 workers, the 100 already on duty are enough. Month 2: 20 000 hours, 125 workers, 25 are hired. Month 3: 28 000 hours, 175 workers, another 50 are hired. The cost is

$$z(MILP) \leq UB = 825\,000.$$

### Dual: the dual bound.

With  $\alpha_t$  free on the demand balances,  $\beta_t \geq 0$  on the hours and  $\gamma_t$  free on the workforce balances, the recipe is:  $\gamma = 0$ , and one hour of work is worth what it really costs,

$$\bar{\beta}_t = \frac{w}{r} = \frac{1500}{160} = \frac{75}{8},$$

so that one pair is worth at most

$$\bar{\alpha}_t = 15 + 4 \cdot \frac{75}{8} = \frac{105}{2} = 52.5.$$

One checks that all the dual columns are satisfied, and

$$LB = \frac{105}{2} \cdot 14\,500 = 761\,250.$$

## Dual of EX 14 on the instance

$$\begin{array}{llllll}
\max & 2500\alpha_1 + 5000\alpha_2 + 7000\alpha_3 & & +100\gamma_1 & & \\
\text{subject to} & \alpha_1 & & -4\beta_1 & & \leq 15 \\
& & \alpha_2 & & -4\beta_2 & \leq 15 \\
& & & \alpha_3 & & -4\beta_3 \leq 15 \\
& -\alpha_1 & +\alpha_2 & & & \leq 3 \\
& & -\alpha_2 & +\alpha_3 & & \leq 3 \\
& & & 160\beta_1 & & +\gamma_1 - \gamma_2 \leq 1500 \\
& & & & 160\beta_2 & +\gamma_2 - \gamma_3 \leq 1500 \\
& & & & & 160\beta_3 & +\gamma_3 \leq 1500 \\
& & & & & & -\gamma_1 \leq 100 \\
& & & & & & -\gamma_2 \leq 100 \\
& & & & & & -\gamma_3 \leq 100 \\
& \alpha_1, & \alpha_2, & \alpha_3 & & & \geq 0 \\
& & & & \beta_1, & \beta_2, & \beta_3 \geq 0 \\
& & & & & & \gamma_1, \gamma_2, \gamma_3 \geq 0
\end{array}$$

## Results.

|             | UB      | LB      | $z(LP)$ | $z(LP^+)$ | $z(MILP)$ |
|-------------|---------|---------|---------|-----------|-----------|
| EX 14 shoes | 825 000 | 761 250 | 773 500 | 773 500   | 774 180   |

The optimal plan produces 3980, 4000 and 6520 pairs with 100, 100 and 163 workers: 1480 pairs are kept in stock at the end of the first month and 480 at the end of the second, and 63 workers are hired once only, in the third month. Producing early costs 3 euros a pair and saves wages: raising the storage cost to 20 euros the inventories drop to 1000 and 0 (optimum 807 500), at 60 euros they disappear altogether and one falls back on the heuristic (825 000).

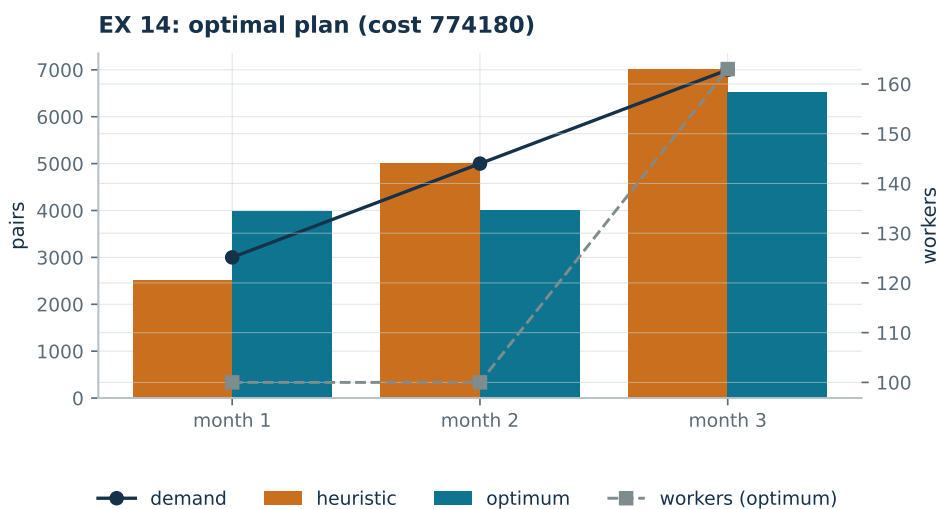


Figure 14: EX 14: the optimal plan of production, stock and workforce, month by month.

## EX 15 — The music school timetable

Script: `python/ex15_timetable.py`

[Notebook in Colab](#)

### EX 15 — weekly timetable

A music school needs to build its weekly timetable. There are two afternoons (Monday and Tuesday), each of two hours. 2 hours of guitar and 2 of violin must be placed. Every day must include lessons of at least two different instruments. Each of the two teachers has one preference, to be respected if possible: the guitar teacher would rather not come on Tuesday, the violin teacher would rather not come on Monday. The goal is to minimise the number of hours that violate a preference.

### The variables

Two families of binary variables:  $x_{dti} = 1$  if at hour  $t$  of day  $d$  instrument  $i$  is taught, and  $y_{di} = 1$  if instrument  $i$  appears on day  $d$ . They are eight columns for the lessons — one per triple day, hour, instrument — and four for the indicators.

### EX 15 on the instance — primal

$$\begin{array}{llllllll}
 \min & x_{112} & +x_{122}+x_{211} & +x_{221} & & & & \\
 \text{subject to} & x_{111} & +x_{121} & +x_{211} & +x_{221} & & & = 2 \\
 & x_{112} & +x_{122} & +x_{212} & +x_{222} & & & = 2 \\
 & x_{111}+x_{112} & & & & & & \leq 1 \\
 & & x_{121}+x_{122} & & & & & \leq 1 \\
 & & & x_{211}+x_{212} & & & & \leq 1 \\
 & & & & x_{221}+x_{222} & & & \leq 1 \\
 & & & & & y_{11}+y_{12} & & \geq 2 \\
 & & & & & & y_{21}+y_{22} & \geq 2 \\
 & x_{111} & & & & -y_{11} & & \leq 0 \\
 & & x_{112} & & & & -y_{12} & \leq 0 \\
 & & & x_{121} & & & -y_{11} & \leq 0 \\
 & & & & x_{122} & & -y_{12} & \leq 0 \\
 & & & & & x_{211} & & -y_{21} \leq 0 \\
 & & & & & & x_{212} & -y_{22} \leq 0 \\
 & & & & & & & x_{221} -y_{21} \leq 0 \\
 & & & & & & & & x_{222} -y_{22} \leq 0 \\
 & -x_{111} & -x_{121} & & & & +y_{11} & \leq 0 \\
 & & -x_{112} & -x_{122} & & & & +y_{12} \leq 0 \\
 & & & & -x_{211} & -x_{221} & & +y_{21} \leq 0 \\
 & & & & & -x_{212} & -x_{222} & +y_{22} \leq 0 \\
 & x_{111}, x_{112}, x_{121}, x_{122}, x_{211}, x_{212}, x_{221}, x_{222} & & & & & & \in \{0,1\} \\
 & & & & & & & & y_{11}, y_{12}, y_{21}, y_{22} \in \{0,1\}
 \end{array}$$

**Description.** The first two rows count the hours of each instrument. The next four say that at most one lesson happens in each slot of the timetable. The two after that are the *variety*: at least two instruments a day. Then come the two directions of the link between lesson and indicator — eight rows for "if there is a lesson the indicator is on" and four for the converse — and at the bottom the two domain rows. The hours to place are  $2 + 2 = 4$  and the slots are  $2 \cdot 2 = 4$ : the two figures coincide, so every slot holds exactly one lesson.

It comes naturally to write only the link  $x_{dti} \leq y_{di}$ , and not its converse. With that direction alone the constraint  $\sum_i y_{di} \geq 2$  is *always* satisfied: it is enough to set  $y_{di} = 1$  without teaching. Solving the model written that way, the solver puts all the guitar on Monday and all the violin on Tuesday — *two* days with a single instrument each — and declares 0 violated preferences, while the true optimum violates 2. The wrong model does not merely allow a bad timetable: it also announces a cost that does not exist.

The link  $x_{dti} \leq y_{di}$  says "if there is a lesson then the indicator is on", not the converse. One also needs

$$y_{di} \leq \sum_{t \in T} x_{dti},$$

that is technique 6.10 (if and only if). It is the same mistake, and the same correction, as in problem 10.3: when a constraint *counts* indicators, the link must hold in both directions.

### A feasible solution.

With the corrected model one builds this timetable by hand (the asterisk marks a violated preference):

|         | hour 1  | hour 2  |
|---------|---------|---------|
| Monday  | guitar  | violin* |
| Tuesday | guitar* | violin  |

One checks: 2 hours of guitar and 2 of violin, two instruments on each day. The variety forces every day to hold both instruments, and every day disappoints one of them: two violated preferences, and one cannot do better.

$$z(MILP) \leq UB = 2.$$

### The lower bound.

All the costs  $c_{dti}$  are 0 or 1, so the objective is a sum of non-negative terms and  $z(MILP) \geq 0$  with no dual needed. The dual of the relaxation confirms that one cannot do better: one variable per family of primal constraints —  $\alpha_i$  free on the hours of each instrument,  $\beta_{dt} \leq 0$  on the slots,  $\gamma_d \geq 0$  on the variety,  $\delta_{dti} \leq 0$  and  $\varepsilon_{di} \leq 0$  on the two directions of the link — and its optimum is 0.

## EX 15 on the instance — dual

$$\begin{array}{llllllllll}
\max & 2\alpha_1 + 2\alpha_2 + \beta_{11} + \beta_{12} + \beta_{21} + \beta_{22} + 2\gamma_1 + 2\gamma_2 & & & & & & & & \\
\text{subject to} & \alpha_1 & +\beta_{11} & & +\delta_{111} & & & & -\epsilon_{11} & \leq 0 \\
& & \alpha_2 + \beta_{11} & & +\delta_{112} & & & & -\epsilon_{12} & \leq 1 \\
& \alpha_1 & & +\beta_{12} & & +\delta_{121} & & & -\epsilon_{11} & \leq 0 \\
& & \alpha_2 & +\beta_{12} & & +\delta_{122} & & & -\epsilon_{12} & \leq 1 \\
& \alpha_1 & & & +\beta_{21} & & +\delta_{211} & & -\epsilon_{21} & \leq 1 \\
& & \alpha_2 & & +\beta_{21} & & +\delta_{212} & & -\epsilon_{22} & \leq 0 \\
& \alpha_1 & & & +\beta_{22} & & +\delta_{221} & & -\epsilon_{21} & \leq 1 \\
& & \alpha_2 & & +\beta_{22} & & +\delta_{222} & & -\epsilon_{22} & \leq 0 \\
& & & \gamma_1 & -\delta_{111} & -\delta_{121} & & & +\epsilon_{11} & \leq 0 \\
& & & \gamma_1 & & -\delta_{112} & -\delta_{122} & & +\epsilon_{12} & \leq 0 \\
& & & & \gamma_2 & & -\delta_{211} & -\delta_{221} & +\epsilon_{21} & \leq 0 \\
& & & & \gamma_2 & & & -\delta_{212} & -\delta_{222} & +\epsilon_{22} \leq 0 \\
& \alpha_1, & \alpha_2 & & & & & & & \geq 0 \\
& & \beta_{11}, \beta_{12}, \beta_{21}, \beta_{22} & & & & & & & \leq 0 \\
& & & \gamma_1, & \gamma_2 & & & & & \geq 0 \\
& & & & \delta_{111}, \delta_{112}, \delta_{121}, \delta_{122}, \delta_{211}, \delta_{212}, \delta_{221}, \delta_{222} & & & & \leq 0 \\
& & & & & & & & \epsilon_{11}, \epsilon_{12}, \epsilon_{21}, \epsilon_{22} & \leq 0
\end{array}$$

**Description.** Every row is a column of the primal: the first eight are the  $x_{dti}$ , where the price of the instrument's hour plus that of the slot, corrected by the two directions of the link, cannot exceed the cost of the slot; the last four are the  $y_{di}$ , where switching the indicator on costs the variety and is refunded by the link.

$$LB = 0, \quad z(LP) = 0, \quad z(LP^+) = 2, \quad z(MILP) = 2.$$

The hand-built timetable was already optimal, but the dual bound does not prove it: between 0 and 2 a gap is left that only branch and bound closes. The strengthened relaxation, which adds  $x \leq 1$  and  $y \leq 1$ , reaches 2 instead and settles it.

Counting says more: every day holds both instruments, and on each day one of the two teachers would rather not be there; the violations are therefore at least two, and  $z(MILP) \geq 2$ . That count, however, uses the integrality of the variables: on the relaxation it does not hold, and indeed  $z(LP) = 0$ . It cannot take the place of  $LB$  in the chain of bounds, where every term must hold for the LP as well.

### What happens if the preferences get tighter.

Two variants make the problem non-trivial.

- If the guitar teacher would rather not teach at hour 2 of Monday either, the only slot left penalty-free for the guitar is hour 1 of Monday, while the hours to place are 2: at least one more violation.

- If at least *three* different instruments a day are required, three instruments would be needed in only two hours. The model is infeasible, and one proves it by counting: Gurobi returns the status **INFEASIBLE**, as expected.

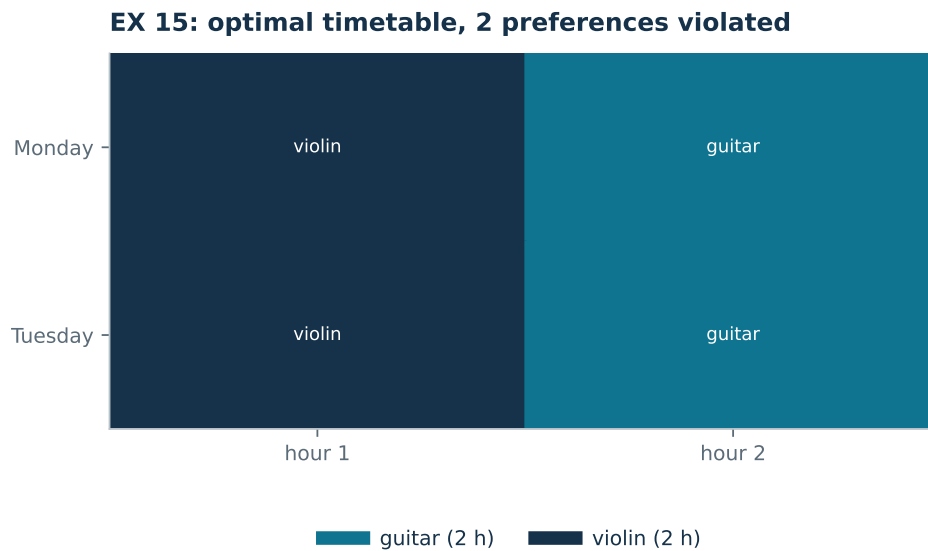


Figure 15: EX 15: the optimal timetable found by the solver, one colour per instrument.

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